

Expanded Blood Pressure Value Edit Integrity Analysis

System, Specialty, Department, Physician, Demographic, and Time-Series Findings - CY2025

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Executive Summary

This report analyzes the 20,357 blood pressure (BP) readings in the THS ambulatory EHR from CY25 that subsequently edited to a different value. The question is not whether individual edits are ever legitimate (they are - device re-checks, transcription fixes, and manual re-measurement all produce genuine corrections) but whether, in aggregate, the population of edits behaves like random correction noise or like a systematic, directional pattern. It does not behave like noise.

This is an analysis of the edit itself, not of BP control or of patient contacts generally. It covers only encounters where a recorded BP value was changed after entry. Earlier versions of this report could not say what share of all THS BP readings this population represents; this update adds a full CY2025 denominator (all BP entries, edited and unedited) pulled directly from Clarity, so system-wide, department-level, and demographic edited-share figures are now reported directly rather than flagged as unanswerable.

Of 20,532 candidate edited-BP records, 20,357 (99.1%) passed data-quality screening (7 dropped for a missing edited value, 168 dropped as physiologically implausible). The analyzed set spans 271 physicians, 40 departments, and 2 specialties (Family Medicine, Internal Medicine). Across every level of aggregation tested - system, specialty, department, age cohort, and individual physician - the same four-part signature repeats: edits are systematically lower than the original reading, edits compress the spread of values toward the center of the range, edits show a stronger preference for round numbers than the original readings did, and the pattern is more pronounced, not less, when the edit is made by someone other than the person who took the original reading.

This update also confirms the dataset captures exactly one editing workflow (an overwrite of the original reading, not a separately charted second measurement), adds edit-magnitude deep dives by age, sex, race, and ethnicity to test whether those factors relate to the size of an edit and not just whether one occurs, and tests whether edit timing (how long after the original reading, and whether it clusters near year-end) predicts edit size. The comment-field / documented-recheck-reason angle raised in review is not included in this version pending a data-quality check on that extract, described in Limitations.

Headline Findings

- Systematic downward bias (H1): mean systolic change of -11.33 mmHg and mean diastolic change of -3.53 mmHg across all 20,357 edits, both significant at $p < 0.001$ (paired t-test and Wilcoxon signed-rank). Effect size for systolic is -0.60 (Cohen's d) - a medium-to-large effect, not a rounding artifact.
- Range compression (H2): the variance of edited systolic values is roughly half that of initial values (variance ratio $1.95\times$, $p < 0.001$); edited readings cluster into a narrower band (178-point range utilized vs. 229 initially).
- Round-number preference emerges upon editing, not before (H3): the terminal-digit chi-square test is not significant for either initial systolic or initial diastolic values, but is significant at $p < 0.001$ for both edited systolic and edited diastolic. Diastolic round-number share nearly jumps from 28.1% to 38.0% after editing.
- Clinical control-rate reclassification, within the edited subset: using the standard ≥ 140 systolic / ≥ 90 diastolic "uncontrolled" threshold used in CMS/NCQA hypertension-control quality measures, 78.2% of these edited encounters were initially uncontrolled; only 20.5% remain uncontrolled after editing. 58.6% of the 20,357 edited

encounters - and 74.9% of the ones that started uncontrolled - were reclassified into the "controlled" category purely as a byproduct of the edit.

- The bias is stronger when the editor is not the original recorder: 68.3% of these edits (13,909 of 20,357) were made by someone other than the person who recorded the original value. This subgroup shows a larger effect size (-0.72 vs. -0.60 across the full edited set) and a much higher reclassification rate - 86.3% vs. 47.7% for same-user edits (Mann-Whitney $U = 40,571,504$, $p < 0.001$).
- NEW - System-wide edited share is small in absolute terms: 0.25% of all 8,233,725 CY2025 BP entries in Family Medicine/Internal Medicine were edited (20,357 of them). Department-level edited share ranges widely, from 11.1% (Department 04) down to roughly 2%, meaning the bias documented above is concentrated in a small, unevenly-distributed slice of total BP volume, not spread evenly across it.
- NEW - Edit magnitude increases with patient age for systolic readings: mean systolic change grows from -6.95 mmHg (<18) to -13.48 mmHg ($80+$), a small-to-medium effect (Cohen's $d = -0.32$ for $80+$ vs. the $18-39$ baseline). Diastolic magnitude moves the opposite direction, shrinking with age - consistent with isolated systolic hypertension becoming more common in older patients clinically, and a reminder that age's relationship to editing behavior is not a single uniform effect.
- NEW - Timing does not explain edit size: how long after the original reading an edit was filed has essentially no relationship to how big or which direction the change was ($r = -0.01$, $p = 0.36$). 88.0% of edits are filed within an hour of the original reading. Separately, Q4 does carry a disproportionate share of edit volume (28.4% vs. an even 25%) for several physicians, but the average edit filed in Q4 is not larger than the rest-of-year average - for every individually flagged physician, the Q4-vs-rest difference in edit size is not statistically significant. Edit volume concentrates near year-end for some physicians; edit size does not.
- NEW - Of the four demographic factors tested as possible motivators, race and ethnicity show no system-wide edit-magnitude signal at the major-group level (White vs. Black or African American mean systolic change: -11.42 vs. -11.10 , not significant), while sex shows a small but statistically significant difference in average edit size (Female -11.54 vs. Male -11.04 , $p = 0.012$, effect size negligible) and a much larger difference in how variable edit size is between the two (Levene $p < 0.001$). Age (above) remains the strongest demographic signal of the four.

1. Purpose and Scope

This analysis applies the same hypothesis-testing framework used in the preliminary half-year manual review (H1 2025, n = 10,019 initial readings) to the full CY25 population, now run as a reproducible notebook against the raw edit-history export rather than a manually built workbook. The underlying question is whether BP edits reflect legitimate clinical correction or a systematic pattern consistent with values being adjusted toward round, "cleaner," or better-controlled numbers after the fact. Because BP control rate is a component of multiple externally reported and reimbursement-linked quality measures (e.g., CMS MIPS BP control, NCQA Controlling High Blood Pressure), a directional edit pattern has downstream implications beyond documentation hygiene and it is reported here to surface that risk, not to assign individual intent; this analysis cannot distinguish a well-intentioned repeat measurement from a value adjusted to hit a target.

The analysis includes a dedicated sub-analysis isolating edits made by someone other than the clinician or staff member who recorded the original reading, since that pattern (one person's number being changed by a second person, often after the encounter) is a materially different workflow and risk profile than a single user correcting their own entry.

A scope note that applies to Sections 3.1–3.8: the source data underlying those sections contains only encounters where a BP value was edited - it is not a sample of, or a stand-in for, all BP readings taken at THS in 2025. Sections 3.9 onward incorporate a separate, full-population denominator extract (all CY2025 BP entries, edited and unedited) so that system, department, and demographic edited-share figures can be reported directly rather than inferred. Every percentage in Sections 3.1–3.8 still describes the edited-encounter subset only, and is labeled as such throughout.

2. Data and Methodology

2.1 Source and Screening

Source file: Edited BP Values, January 1 – December 31, 2025 (THS ambulatory EHR edit-history export). Each row represents one encounter where the recorded systolic/diastolic value was changed at least once after initial entry, with the initial and final (edited) values, the user who took/entered each, and the elapsed time between the original filing and the edit.

Table 1. Data screening summary.

Screening Step	Rows
Rows in source file	20,532
Dropped - missing edited reading	7
Dropped - physiologically implausible value	168
Rows analyzed	20,357

The analyzed population covers 271 unique physicians, 40 departments, and 2 specialties (Family Medicine and Internal Medicine - the only two specialties represented in this extract).

2.2 Hypotheses Tested

- H1 - Systematic Bias: paired t-test $p < 0.05$, $|\text{effect size}| > 0.20$, and mean difference $\neq 0$.
- H2 - Range Compression: F-test $p < 0.05$ and variance ratio > 1.50 , with the edited range narrower than the initial range.
- H3 - Digit Preference: round-number share $> 30\%$ and/or a significant terminal-digit chi-square test ($p < 0.05$).
- H4 - Department-Specific Bias: ANOVA across departments $p < 0.05$ and Levene's test $p < 0.05$ (variance itself differs by department).
- H4b - Age-Cohort-Specific Bias: same test structure as H4, applied across five age cohorts (<18 , $18-39$, $40-59$, $60-79$, $80+$, derived from single-year age at the time of the reading).

Each scope (system / specialty / department / physician / age cohort / editor) is scored against these criteria and assigned a Flag Level:

- CRITICAL (≥ 8 of 13 criteria met)
- HIGH (≥ 5)
- MODERATE (≥ 3)
- LOW (≥ 1)
- CLEAN (0)
- INSUFFICIENT ($n < 10$, too small to test reliably).

2.3 Clinical Control-Rate Cross-Check

In addition to the notebook's statistical flagging criteria, this report includes a supplemental, independently computed metric: the share of encounters classified as "uncontrolled" (systolic ≥ 140 mmHg or diastolic ≥ 90 mmHg - the threshold used by CMS and NCQA hypertension-control quality measures) before versus after the edit. This metric is not one of the H1–H4b flagging criteria; it is included because it translates the statistical findings into a clinically and administratively legible quantity.

2.4 Denominator and Demographic Data Source (New)

Sections 3.9–3.15 use a second, independent extract: every BP flowsheet entry recorded in CY2025 (FLO_MEAS_ID = 5 in IP_FLWSHT_MEAS), not just the ones later edited, joined to PATIENT / PATIENT_RACE / ZC_SEX / ZC_PATIENT_RACE / ZC_ETHNIC_GROUP for demographics. Per Clarity guidance from Data Engineering, ENTRY_TIME (when the reading was physically entered into Epic) is used as the CY2025 boundary, not RECORDED_TIME (when the reading was observed). The raw pull returned 8,233,726 rows; a defensive dedup keyed on (encounter, recorded time, entry time, value, entry user) - the combination that uniquely identifies one flowsheet reading - removed exactly 1 duplicate row, confirming the extract is effectively clean rather than fanned out by a lookup join, leaving 8,233,725 rows as the system-wide denominator used throughout Sections 3.9–3.15.

This denominator identifies who entered each reading (Entry_User_Name), not the physician of record for the encounter (Enc_Provider, used throughout Sections 3.1–3.8). The two are frequently different people - most entrants in this dataset are nursing/MA staff or the automated device interface, not physicians - so Section 3.12 (entrant-level) and Sections 3.1–3.8 (physician-level) answer different questions and are not directly comparable row-for-row. A join from this denominator to the physician of record has not yet been obtained; see Limitations.

3. Results

3.1 System-Level Findings

"System-level" here means all 20,357 edited encounters pooled together, not all THS ambulatory encounters - see the scope note in Section 1. Within that edited population, all four hypotheses (H1, H2, H3, H4/H4b) are confirmed, and the pooled edited population flags CRITICAL (11 of 13 criteria met).

Table 2. System-wide summary statistics, CY2025.

Metric	Value
Encounters analyzed	20,357
Mean systolic change (edited – initial)	–11.33 mmHg ($p < 0.001$)
Mean diastolic change (edited – initial)	–3.53 mmHg ($p < 0.001$)
Effect size, systolic / diastolic (Cohen's d)	–0.598 / –0.265
Variance ratio, systolic / diastolic (initial:edited)	$1.95\times / 1.67\times$ ($p < 0.001$)
Value range utilized, initial → edited	229 → 178 points
Round-number share, all readings	28.7%
Terminal-digit chi-square	$\chi^2 = 6,146.8$, $p < 0.001$
Most common initial BP (count)	140/80 ($n = 319$)
Most common edited BP (count)	130/80 ($n = 577$)
ANOVA across departments (Systolic Δ)	$F = 70.69$, $p < 0.001$
Levene's test across departments (variance)	$F = 18.55$, $p < 0.001$
ANOVA across age cohorts (Systolic Δ)	$F = 79.82$, $p < 0.001$
Levene's test across age cohorts (variance)	$F = 30.52$, $p < 0.001$
Mann-Whitney, $ \Delta\text{systolic} $ different- vs. same-user editor	$U = 40,571,504$, $p < 0.001$
Flag	CRITICAL (11/13)

Two things stand out beyond statistical significance. First, the most common edited value (130/80, $n = 577$) is reached almost twice as often as the most common initial value (140/80, $n = 319$) - edits are not just moving values down, they are converging on a small set of "clean" numbers. Second, both the department-level and age-cohort-level Levene's tests are significant, meaning the degree of variability in editing behavior - not just the average - differs meaningfully by department and by patient age.

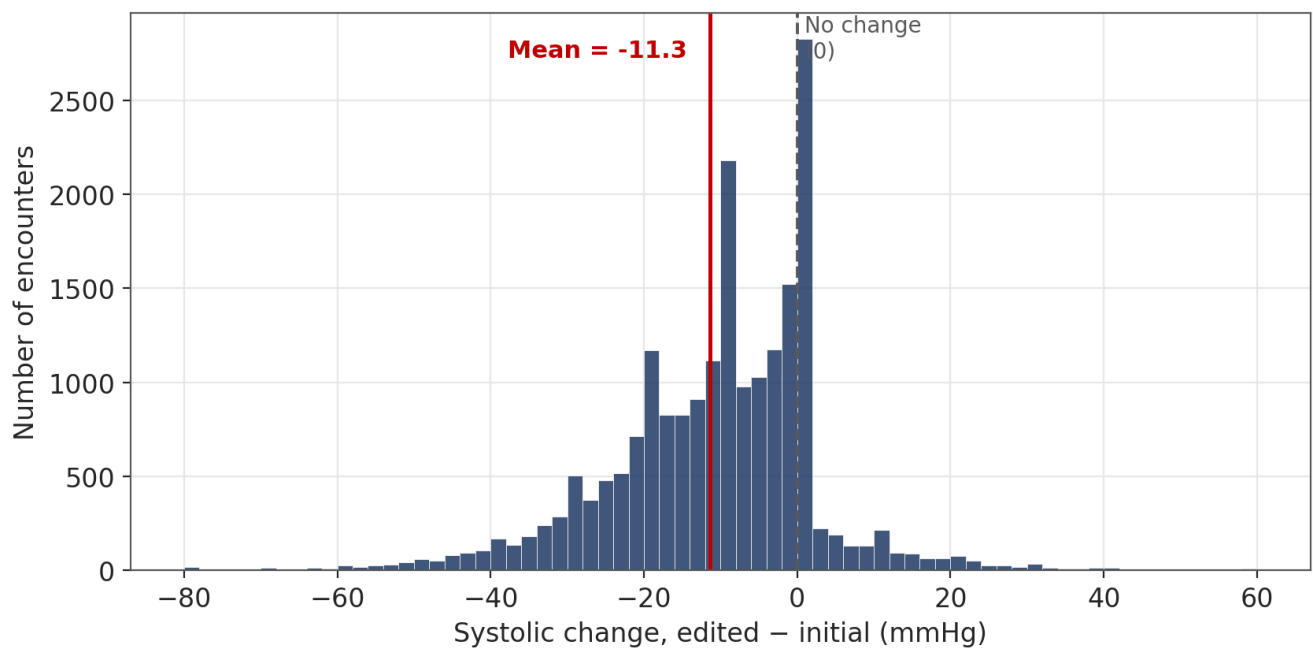


Figure 1. Distribution of the per-encounter systolic edit (edited - initial), all 20,334 plausibility-screened edited encounters. The distribution sits almost entirely left of zero and is not symmetric around it; the sharp spike at 0 partly reflects encounters where only the diastolic value was changed.

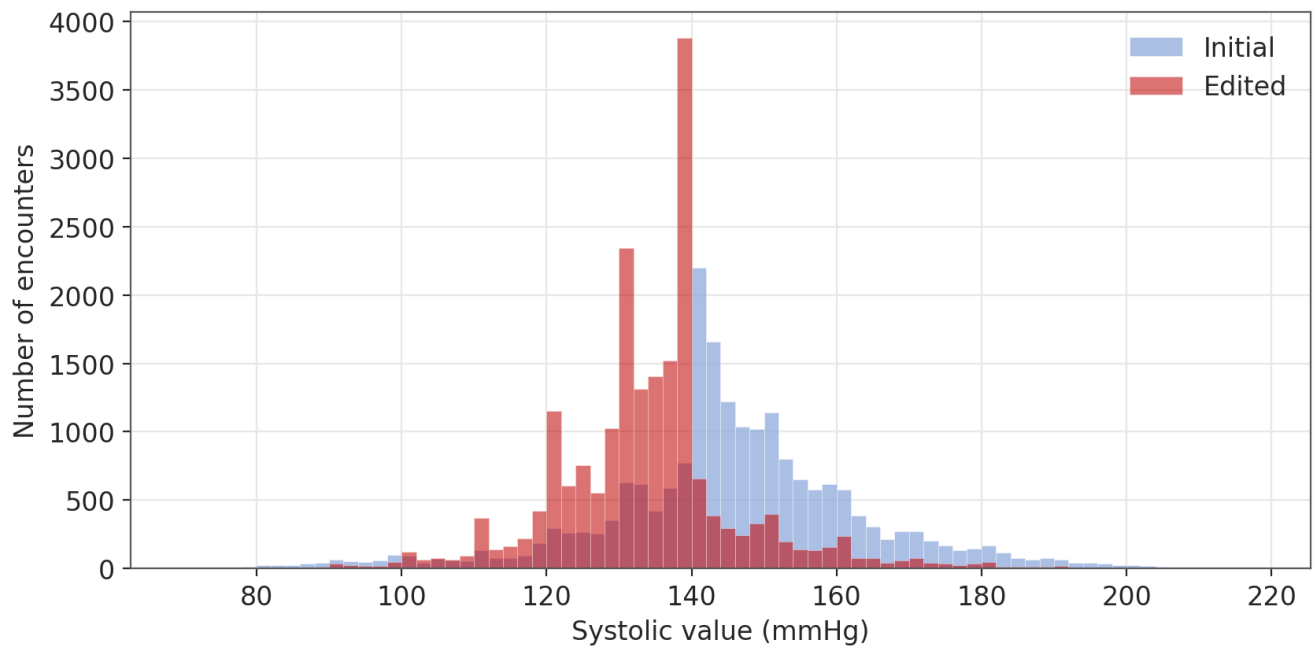


Figure 2. Initial (blue) vs. edited (red) systolic value distributions overlaid. The initial distribution is broad and centered in the mid-140s; the edited distribution is sharply peaked at 130 and 140 - visual evidence of both the downward shift (H1) and the range compression / round-number convergence (H2, H3).

3.2 Clinical Control-Rate Reclassification (Edited Subset Only)

Applying the $\geq 140/90$ "uncontrolled" threshold used in CMS/NCQA quality reporting to the initial and edited values, within the population of encounters that received a BP edit, shows a large shift in apparent BP control that is attributable entirely to the edit itself. This is not a system-wide BP control rate for THS - it describes only the 20,334 edited encounters in this dataset:

Table 3. Uncontrolled-BP reclassification, initial vs. edited value, split by editor type. (N reflects the plausibility-screened subset used for this cross-check, 20,334 of 20,357 rows.)

Population	N	% Uncontrolled, Initial	% Uncontrolled, Edited	Reclassified to Controlled
All edited encounters	20,334	78.7%	20.5%	12,010 (59.1% of all; 75.1% of initially uncontrolled)
Edited by different user	13,902	81.6%	11.6%	9,792 (70.4% of subgroup; 86.3% of initially uncontrolled)
Edited by same user	6,432	72.3%	39.8%	2,218 (34.5% of subgroup; 47.7% of initially uncontrolled)

In plain terms: among encounters that had a BP value edited, roughly 4 in 5 started in "uncontrolled" territory; after editing, only 1 in 5 remain there. Same-user edits still show a large shift, but different-user edits are nearly twice as likely to flip a patient's status from uncontrolled to controlled (86.3% vs. 47.7% of initially-uncontrolled cases).

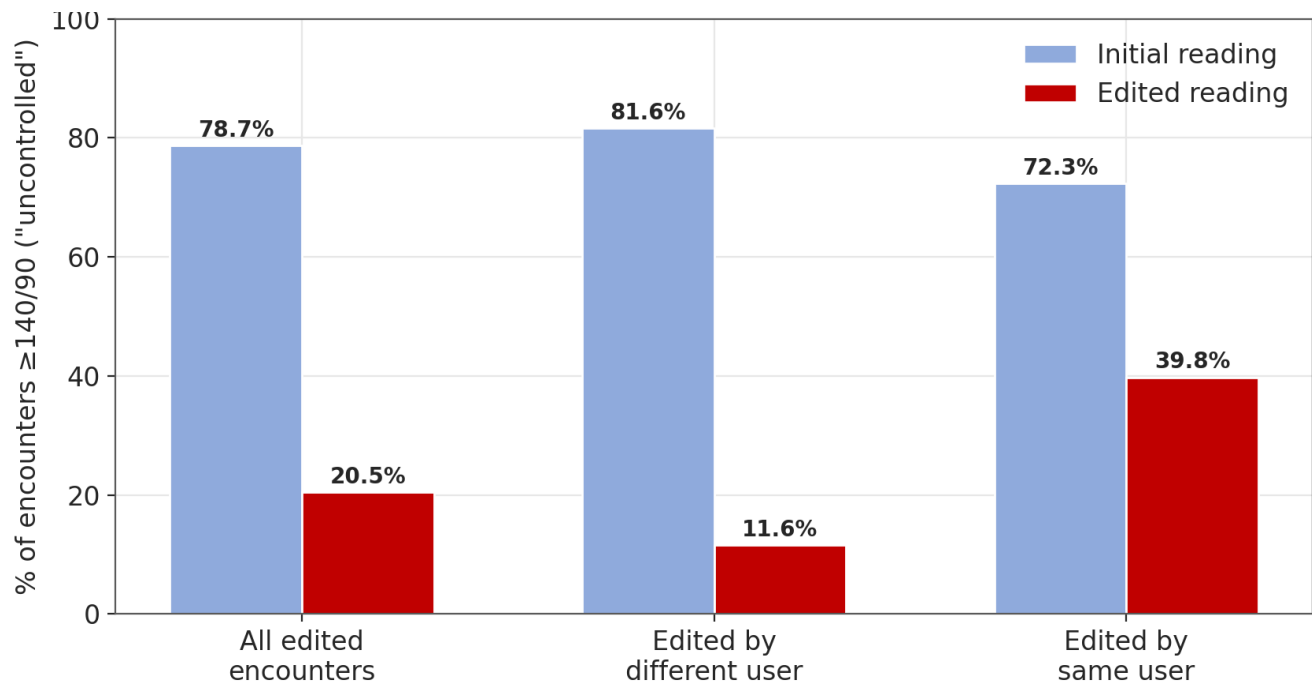


Figure 3. Share of edited encounters classified "uncontrolled" (≥ 140 systolic or ≥ 90 diastolic), initial vs. edited reading, for all edited encounters and split by editor type.

3.3 Specialty-Level Findings

Table 4. Specialty-level summary. Both specialties flag **CRITICAL**; Family Medicine shows the larger systolic effect, Internal Medicine the larger diastolic effect and the higher round-number share.

Specialty	N	% of Total	Effect Size (Sys/Dia)	Mean Δ Sys / Dia	Round-Number %	Flag
Family Medicine	14,506	71.3%	-0.639 / -0.228	-12.49 / -3.19	27.1%	CRITICAL
Internal Medicine	5,851	28.7%	-0.519 / -0.382	-8.46 / -4.38	32.6%	CRITICAL

Specialty-Level Findings — Family Medicine vs. Internal Medicine (Edited-Encounter Subset)

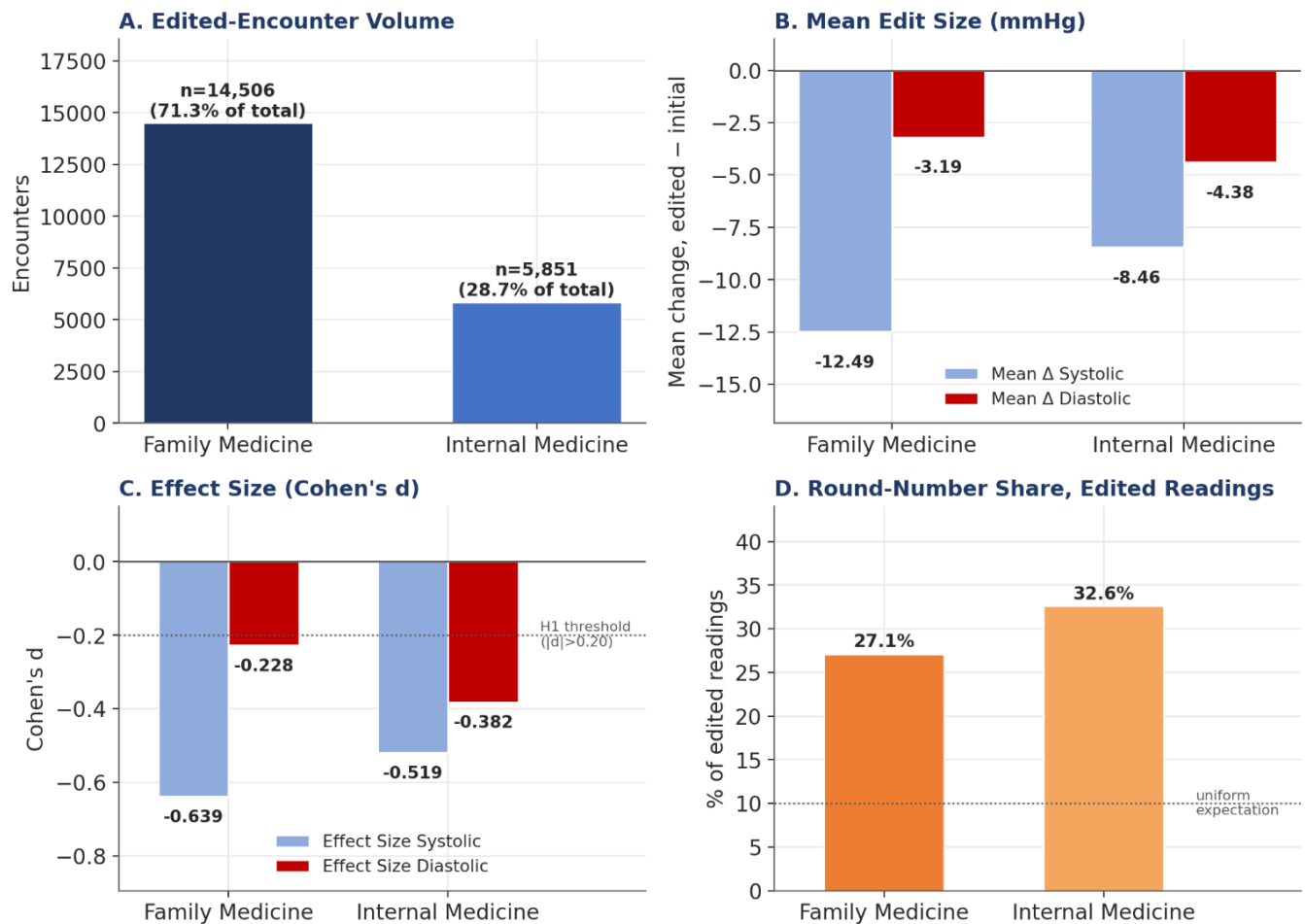


Figure 4. Specialty comparison. Panel A: edited-encounter volume (Family Medicine 71.3% vs. Internal Medicine 28.7%). Panel B: mean systolic/diastolic edit size. Panel C: effect size (Cohen's d) against the H1 threshold (dotted line). Panel D: round-number share against the 10% uniform-digit baseline.

3.4 Department-Level Findings

34 of 40 departments have adequate volume ($n \geq 12$) to evaluate; the remaining 6 are small occupational-health clinics with 1–8 encounters and are marked INSUFFICIENT. Of the 34 evaluable departments, 25 flag CRITICAL, 8 flag HIGH, and 1 flags LOW - no department with meaningful volume comes back CLEAN.

Table 5. All 34 evaluable departments, sorted by criteria met.

Department	N	Crit. Met	Flag	Mean Δ Sys	Mean Δ Dia	Round %
Department 01	1,017	12	CRITICAL	−7.86	−3.44	36.5%
Department 02	757	12	CRITICAL	−20.40	−5.24	23.0%
Department 03	577	12	CRITICAL	−8.30	−3.40	45.8%
Department 04	2,349	11	CRITICAL	−17.44	−4.39	26.9%
Department 05	2,244	11	CRITICAL	−8.70	−7.08	29.0%
Department 06	1,080	11	CRITICAL	−11.68	−4.89	28.1%
Department 07	552	11	CRITICAL	−13.41	−3.67	21.4%
Department 08	238	11	CRITICAL	−16.64	−4.36	23.5%
Department 09	2,214	10	CRITICAL	−16.43	−1.89	23.7%
Department 10	1,511	10	CRITICAL	−6.39	−2.70	35.1%
Department 11	792	10	CRITICAL	−12.05	−1.60	28.5%
Department 12	374	10	CRITICAL	−7.18	−3.39	21.4%
Department 13	346	10	CRITICAL	−6.51	−10.03	40.2%
Department 14	338	10	CRITICAL	−6.59	−2.50	34.3%
Department 15	284	10	CRITICAL	−8.05	−4.52	24.3%
Department 16	48	10	CRITICAL	−12.62	−4.21	27.1%
Department 17	737	9	CRITICAL	−15.13	−2.05	23.2%
Department 18	506	9	CRITICAL	−10.37	−2.35	24.3%
Department 19	478	9	CRITICAL	−6.30	−4.24	27.4%
Department 20	342	9	CRITICAL	−7.95	−4.61	40.1%
Department 21	1,017	8	CRITICAL	−12.18	−1.51	20.6%
Department 22	172	8	CRITICAL	−8.23	−4.49	23.3%
Department 23	86	8	CRITICAL	−5.91	−4.31	17.4%
Department 24	63	8	CRITICAL	−5.22	0.14	52.4%
Department 25	531	7	HIGH	−8.51	−1.03	22.0%
Department 26	414	7	HIGH	−6.18	−1.10	28.5%
Department 27	234	7	HIGH	−13.55	−1.09	32.1%
Department 28	169	7	HIGH	−10.21	−1.33	23.7%
Department 29	165	7	HIGH	−4.04	−0.45	36.4%
Department 30	36	7	HIGH	−7.69	−3.42	16.7%
Department 32	275	6	HIGH	−2.67	−0.20	28.0%
Department 33	240	5	HIGH	−2.12	−0.91	67.5%
Department 31	143	5	HIGH	−8.61	−2.43	25.2%
Department 34	12	2	LOW	0.58	3.00	50.0%

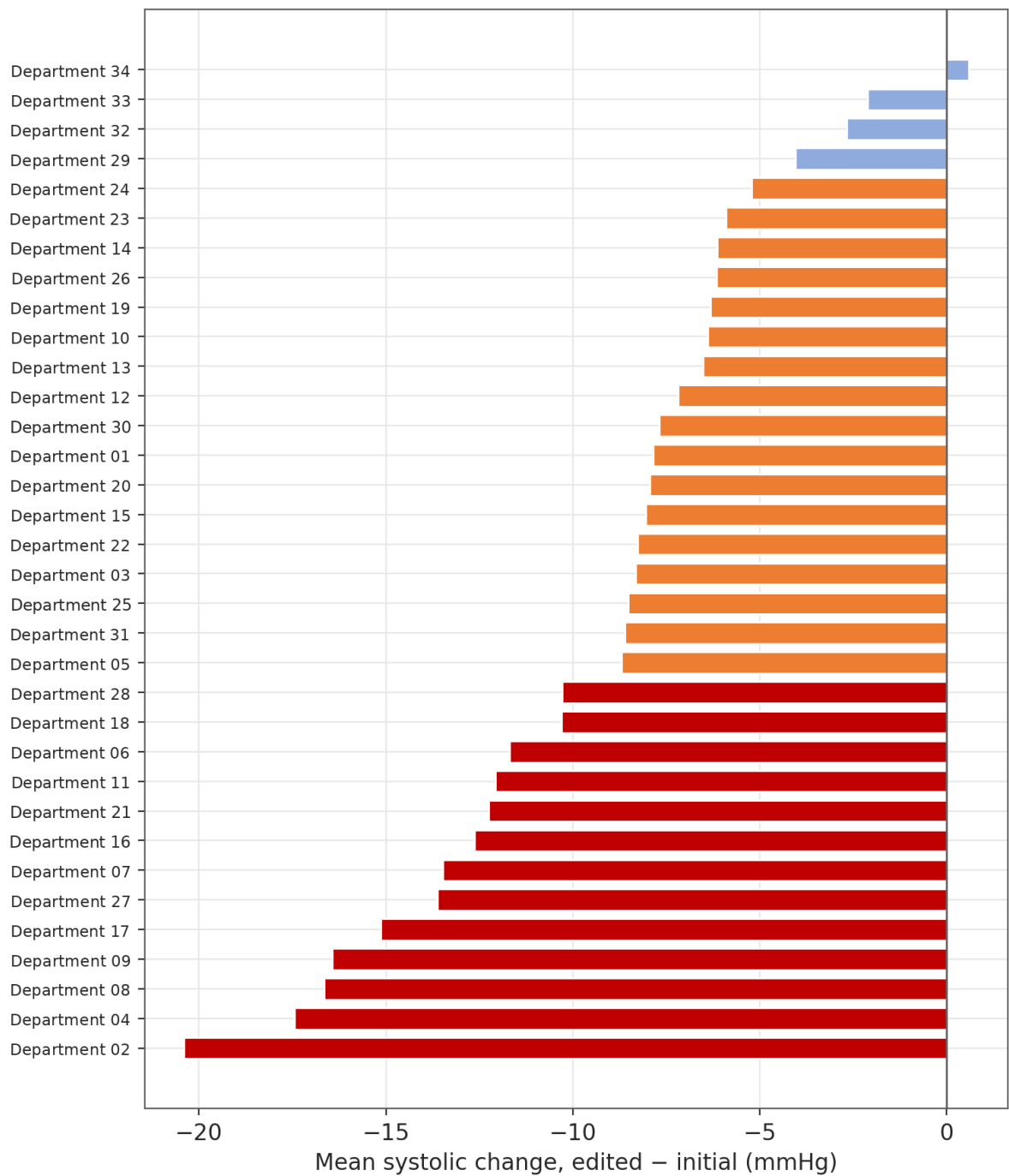


Figure 5. Mean systolic edit by department ($n \geq 12$), sorted low to high. Red = CRITICAL-range shift (≤ -10 mmHg mean), orange = moderate shift (-5 to -10), blue = smaller shift.

3.5 Physician-Level Findings

Of 271 physicians with at least one edited BP record, 107 have fewer than 10 encounters and are marked INSUFFICIENT. Among the remaining 164 physicians with adequate volume, 65 (39.6%) flag CRITICAL and 47 (28.7%) flag HIGH - together, 68.3% of evaluable physicians flag CRITICAL or HIGH.

Table 6. Physician flag-level distribution.

Flag Level	Physicians	% of Evaluable (n=164)
CRITICAL	65	39.6%
HIGH	47	28.7%
MODERATE	26	15.9%
LOW	20	12.2%
CLEAN	6	3.7%
(INSUFFICIENT, n<10 - excluded above)	107	–

The 20 physicians below (minimum n = 30) meet the most flagging criteria.

Table 7. Top 20 physicians by criteria met (min. n = 30), by encounter provider of record.

Physician (encounter provider)	N	Crit. Met	Flag	Mean Δ Systolic
Physician 01	137	13	CRITICAL	–20.97
Physician 02	285	13	CRITICAL	–9.75
Physician 03	105	12	CRITICAL	–21.02
Physician 04	414	12	CRITICAL	–21.01
Physician 05	1,156	12	CRITICAL	–19.58
Physician 06	487	12	CRITICAL	–17.57
Physician 07	290	12	CRITICAL	–13.97
Physician 08	565	12	CRITICAL	–12.11
Physician 09	269	12	CRITICAL	–11.42
Physician 10	104	12	CRITICAL	–7.16
Physician 11	113	11	CRITICAL	–25.73
Physician 12	343	11	CRITICAL	–19.67
Physician 13	170	11	CRITICAL	–17.50
Physician 14	59	11	CRITICAL	–17.20
Physician 15	207	11	CRITICAL	–16.01
Physician 16	232	11	CRITICAL	–15.81
Physician 17	439	11	CRITICAL	–13.80
Physician 18	274	11	CRITICAL	–11.81
Physician 19	1,761	11	CRITICAL	–8.34
Physician 20	302	11	CRITICAL	–6.96

3.6 Digit Preference (H3)

Round-number preference is markedly stronger after editing than before it, and the shift is larger for diastolic than systolic values.

Table 8. Terminal-digit distribution test, initial vs. edited readings.

Reading	N	χ^2 (terminal digit)	p	Round-Number %	Digit-Preference Flag
Initial Systolic	20,357	6,146.8	< 0.001	28.7%	No
Edited Systolic	20,357	11,575.3	< 0.001	30.2%	Yes
Initial Diastolic	20,357	5,992.8	< 0.001	28.1%	No
Edited Diastolic	20,357	16,591.2	< 0.001	38.0%	Yes

A secondary pattern: edited systolic values show a pronounced excess of readings ending in 8 (23.0% of edited systolic values vs. 13.3% of initial), on top of the more familiar excess ending in 0 (24.6% vs. 22.6%). Edited diastolic values show the same secondary pattern (18.6% vs. 13.8% ending in 8), consistent with edits converging on a small set of memorable readings - 130/80, 138/80, 120/80 - rather than pure rounding-to-nearest-10.

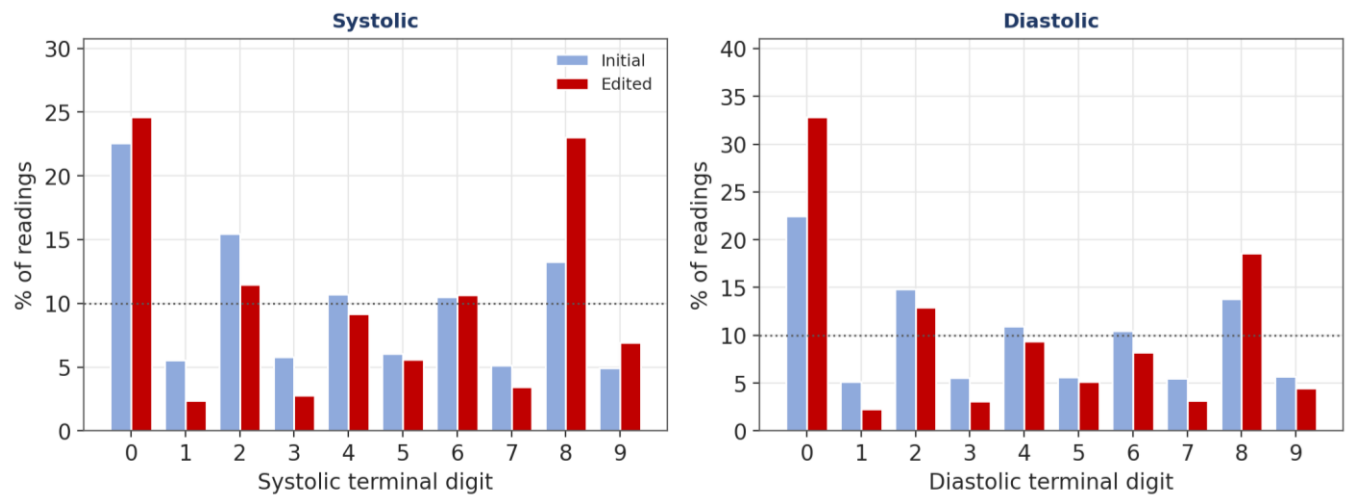


Figure 6. Terminal-digit distribution (last digit of each reading), initial vs. edited, systolic and diastolic. A uniform, unbiased distribution would sit at 10% for every digit (dotted line).

3.7 Sub-Analysis: Edits by a Different User

68.3% of all edits in this dataset (13,909 of 20,357) were made by a user other than the one who took and originally entered the reading.

Table 9. Different-user edit subgroup, system and by specialty.

Scope	N	Effect Size (Sys/Dia)	Mean Δ Sys / Dia	Round-Number %	Flag
System (different-user edits)	13,909	-0.717 / -0.372	-12.22 / -4.49	29.6%	CRITICAL
Family Medicine (different-user)	8,812	-0.821 / -0.345	-14.18 / -4.29	28.5%	CRITICAL
Internal Medicine (different-user)	5,097	-0.567 / -0.435	-8.84 / -4.85	31.4%	CRITICAL

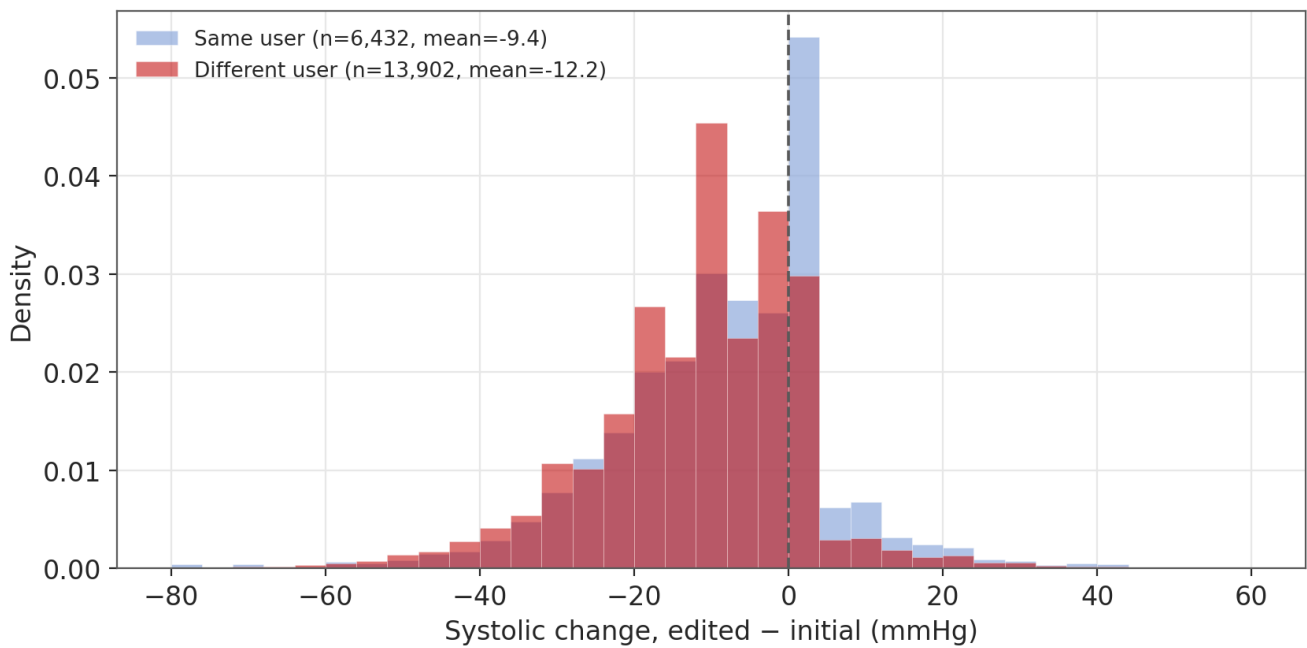


Figure 7. Distribution of per-encounter systolic edit size, same-user edits vs. different-user edits (density-normalized).

Looking at who is doing the editing, 298 distinct users appear as an "Edited_Entry_User" for at least one different-user edit.

Table 10. Top 15 editing users by criteria met (min. $n = 30$).

Editing User (Edited_Entry_User)	N	Crit. Met	Flag	Mean Δ Systolic
Physician 02	269	13	CRITICAL	-10.07
Physician 01	130	13	CRITICAL	-22.14
Physician 05	1,122	12	CRITICAL	-20.10
Physician 08	543	12	CRITICAL	-12.42
Physician 06	463	12	CRITICAL	-18.31
Physician 04	406	12	CRITICAL	-21.38
Physician 07	260	12	CRITICAL	-15.18
Physician 09	237	12	CRITICAL	-12.62
Physician 10	90	12	CRITICAL	-8.09
Physician 03	82	12	CRITICAL	-22.78
Staff 02	37	12	CRITICAL	-13.43
Physician 19	1,724	11	CRITICAL	-8.49
Physician 21	725	11	CRITICAL	-5.36
Physician 17	335	11	CRITICAL	-12.76
Physician 12	329	11	CRITICAL	-20.24

Several names appear on both Table 7 and Table 10 - e.g., Physician 02, Physician 04, Physician 08, Physician 06 - meaning physician-level and editor-level flags are not fully independent signals.

Table 11. Top 10 original-recorder $\rightarrow$ editor pairs by edit volume.

Original Recorder	Editing User	Edits	Mean Δ Sys	Mean Δ Dia	Round %
Staff 03	Physician 19	543	-10.38	-8.96	16.8%
Staff 04	Physician 19	509	-5.92	-8.69	15.7%
Staff 05	Physician 08	477	-12.75	-6.56	22.9%
Staff 06	Physician 21	466	-4.17	-3.82	42.9%
Staff 07	Physician 22	452	-17.51	-2.85	22.8%
Staff 08	Physician 05	445	-20.70	-5.91	38.4%
Staff 09	Physician 05	444	-20.66	-5.68	19.1%
Staff 10	Physician 23	424	-13.58	-1.29	31.6%
Staff 11	Physician 04	337	-21.44	-4.91	22.6%
Staff 12	Staff 13	317	-13.92	-1.58	23.3%

3.8 Age-Cohort Findings

Table 12. Findings by age cohort. Every cohort flags HIGH or worse; the 80+ cohort shows the largest average downward shift.

Age Cohort	N	Crit. Met	Flag	Mean Δ Systolic
60-79	9,675	10	CRITICAL	-12.29
40-59	5,348	11	CRITICAL	-9.34
80+	3,324	11	CRITICAL	-13.48
18-39	1,805	10	CRITICAL	-8.63
<18	205	7	HIGH	-6.95

3.9 Scope Confirmation: Which Edit Workflows Does This Dataset Capture?

In review, Physician 24 raised a scope question central to interpreting every finding above: BP edits in Epic could in principle be recorded two ways - as a second, separately charted reading for the same encounter, or as a direct overwrite of the initial value. If this dataset mixed the two, "edit" would mean different things in different rows.

Every PAT_ENC_CSN_ID in the 20,357-row analyzed set was checked for more than one edit row: 0 encounters have more than one. This confirms the dataset captures exactly one workflow - a direct overwrite of the original value (one initial reading, one edited reading, per encounter) - and not a second, separately-documented reading. A workflow where staff instead chart a second BP as a fresh entry (rather than editing the first) would not appear as an "edit" at all and is structurally invisible to this extract; it can only be assessed with a separate query against all flowsheet entries per encounter, not the edit-history export used here.

3.10 System-Wide and Departmental Edited-Share (Denominator Analysis)

Using the full CY2025 denominator described in Section 2.4, 20,357 of 8,233,725 total BP entries in Family Medicine/Internal Medicine were edited system-wide - 0.25%. This answers the question raised in review of what share of all BP readings are edited: in absolute terms, small; but as Sections 3.1–3.9 show, the 0.25% that are edited behave in a systematic, non-random way, and that share is not evenly spread across departments.

Table 13. Department-level edited share (departments captured in this denominator run, sorted by edited-share descending; not confirmed to include all 40 departments in Table 5).

Department	Total BP Entries	Edited Count	% Edited
Department 04	21,135	2,349	11.11%
Department 05	20,778	2,244	10.80%
Department 13	3,361	346	10.29%
Department 09	21,544	2,214	10.28%
Department 02	8,184	757	9.25%
Department 07	7,477	552	7.38%
Department 06	16,660	1,080	6.48%
Department 01	17,136	1,017	5.93%
Department 11	17,302	792	4.58%
Department 10	43,180	1,511	3.50%
Department 03	19,885	577	2.90%
Department 17	25,775	737	2.86%
Department 18	17,995	506	2.81%
Department 25	19,351	531	2.74%
Department 21	37,516	1,017	2.71%
Department 27	9,102	234	2.57%
Department 19	19,926	478	2.40%
Department 14	14,123	338	2.39%
Department 08	10,412	238	2.29%

Read together with Table 5: departments with the highest edited-share (Department 04, Department 05, Department 09, Department 02) are largely the same departments that flag CRITICAL on edit magnitude, meaning these departments are unusual on both dimensions - more of their BP volume gets edited, and the edits that happen there are also more directional.

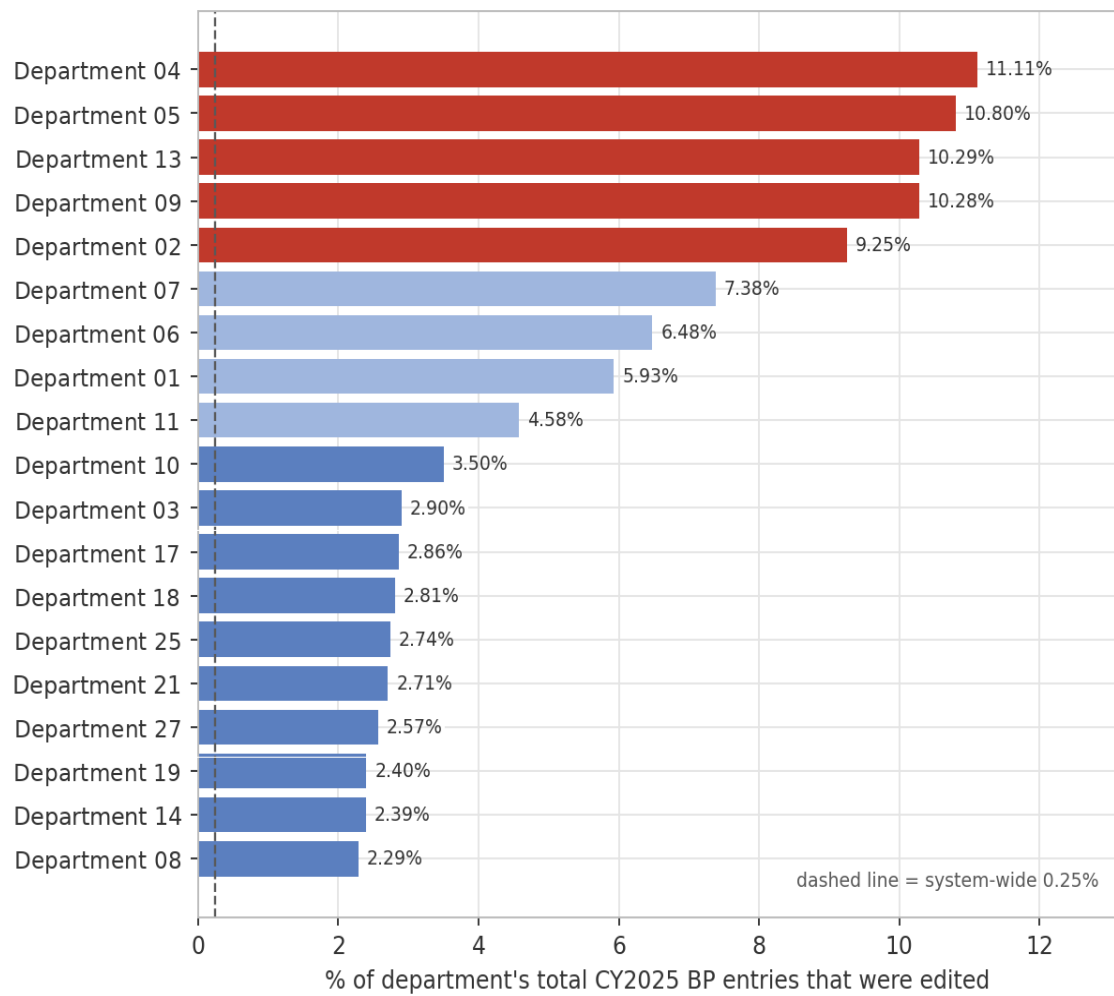


Figure 8. Department-level edited share, sorted descending. Red = highest tier ($\geq 9\%$), blue-scale = lower tiers. Dashed line marks the system-wide 0.25% rate.

3.11 Demographic Edited-Share: Sex, Race, and Ethnicity

The same denominator supports a volume-only breakdown by patient sex, race, and ethnicity - what share of each group's BP entries get edited at all. (Section 3.14 below asks the different question of whether the edits that do happen are bigger for some groups than others.)

Table 14. Sex-level edited share.

Sex	Total BP Entries	Edited Count	% Edited
Female	4,207,789	11,643	0.28%
Male	4,025,812	8,668	0.22%
Unknown	106	0	0.00%

Table 15. Race-level edited share.

Race	Total BP Entries	Edited Count	% Edited
Korean	735	8	1.09%
Filipino	2,292	17	0.74%
Vietnamese	1,673	11	0.66%
American Indian or Alaska Native	2,460	16	0.65%
Chinese	3,234	20	0.62%
Unknown	1,656	8	0.48%
American Indian	7,161	31	0.43%
Other Pacific Islander	1,186	5	0.42%
Asian Indian	17,050	70	0.41%
Decline to Answer	37,292	132	0.35%
Japanese	2,559	8	0.31%
Other Asian	34,258	107	0.31%
White	5,969,417	14,850	0.25%
Black or African American	1,866,601	4,590	0.25%
Native Hawaiian or Other Pacific Islander	7,441	17	0.23%
Other	275,458	419	0.15%
Native Hawaiian	1,984	1	0.05%
Guamanian	303	0	0.00%
Samoan	92	0	0.00%

Small-N groups (Korean, Native Hawaiian, Guamanian, Samoan, etc.) should be read as directional, not conclusive - a handful of edits swings the percentage sharply at this volume.

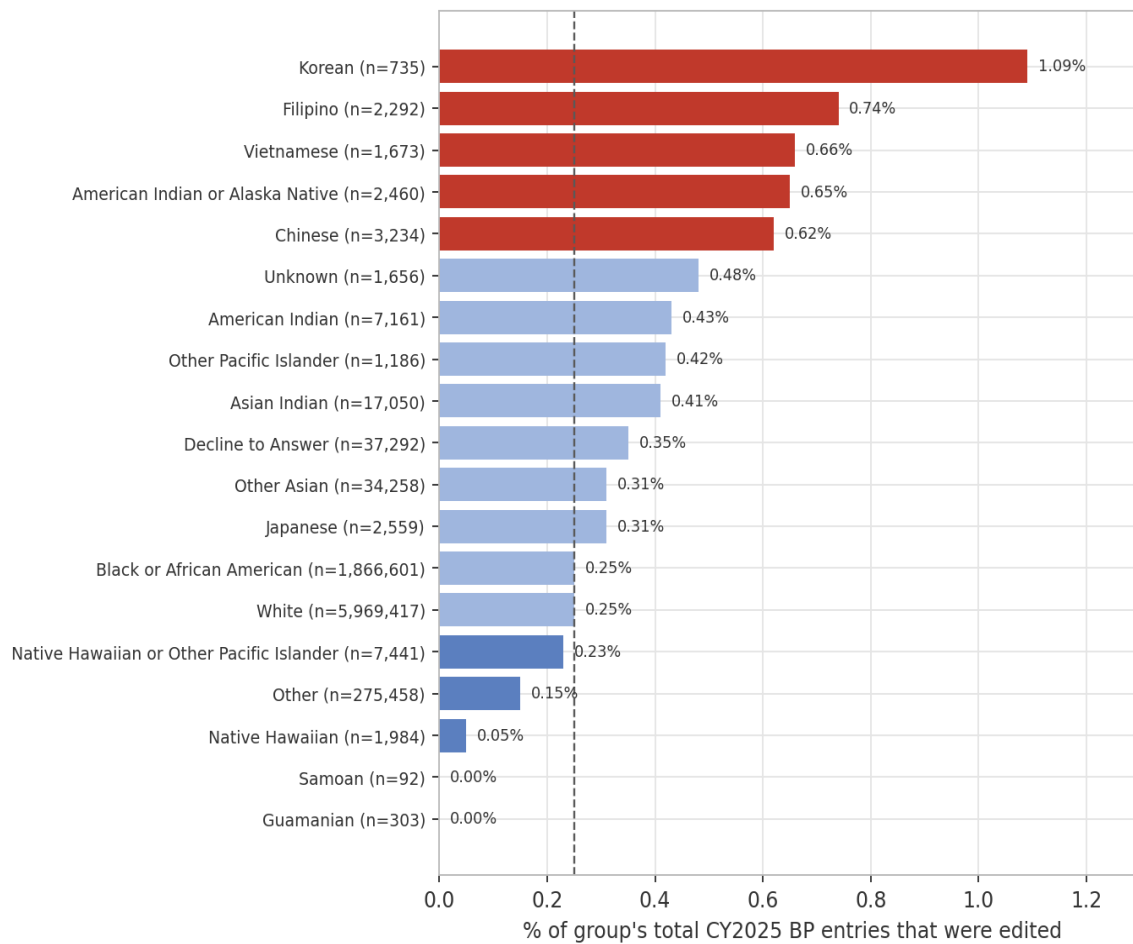


Figure 9. Race-level edited share, with group size (N) labeled since several of the highest-percentage groups are very small. Dashed line marks the system-wide 0.25% rate.

Table 16. Ethnicity-level edited share.

Ethnicity	Total BP Entries	Edited Count	% Edited
Cuban	1,030	7	0.68%
Decline to Answer	43,852	185	0.42%
Unknown	135,337	475	0.35%
Not Hispanic, Latino/a, or Spanish origin	7,656,940	18,898	0.25%
Other Hispanic, Latino/a, or Spanish origin	44,276	107	0.24%
Hispanic or Latino	289,619	590	0.20%
Puerto Rican	13,049	18	0.14%
Mexican, Mexican American, or Chicano/a	48,945	31	0.06%
Ashkenazi	133	0	0.00%

The dominant groups by volume - White, Black or African American (race) and Not Hispanic/Latino (ethnicity) - sit at almost exactly the system-wide 0.25% rate. The groups furthest from that rate (Korean at 1.09%, Cuban at 0.68%) are all small-N, so this table does not by itself establish a demographic edited-share disparity at the largest population groups; it flags specific small groups worth a second look with more data.

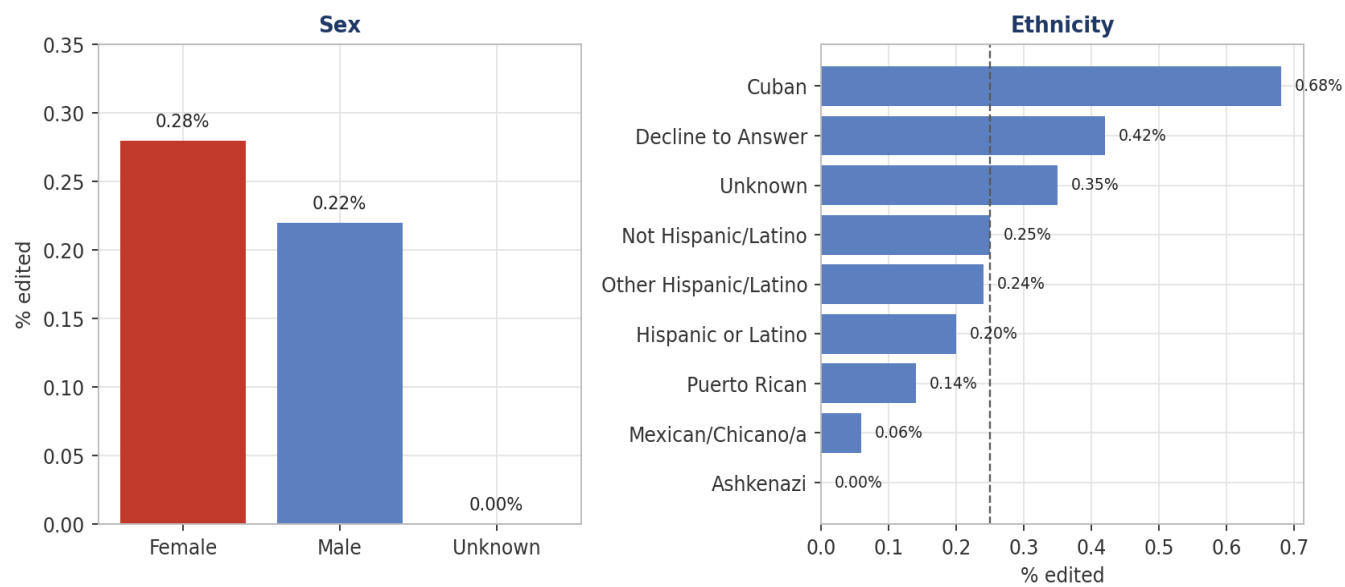


Figure 10. Sex- and ethnicity-level edited share. Dashed line in the ethnicity panel marks the system-wide 0.25% rate.

3.12 Entrant-Level Breakdown

Section 2.4 draws the distinction between Entry_User_Name (who physically entered the reading - usually nursing/MA staff or the automated device interface) and Enc_Provider (the physician of record, used throughout Sections 3.1–3.8). This section reports edited share by entrant. 563 of 566 distinct people who originally entered an edited reading were matched to a denominator record by name.

Table 17. Entrant-level edited share, top 19 by % edited among entrants with a denominator match.

Entrant (Entry_User_Name)	Total BP Entries	Edited Count	% Edited
Staff 14	6	5	83.33%
Staff 03	760	547	71.97%
Staff 09	679	447	65.83%
Staff 15	38	20	52.63%
Staff 16	1,389	466	33.55%
Staff 04	1,593	516	32.39%
Staff 12	1,092	343	31.41%
Staff 17	81	25	30.86%
Staff 08	1,758	517	29.41%
Staff 18	2,558	717	28.03%
Staff 07	3,179	801	25.20%
Staff 19	4	1	25.00%
Staff 10	1,811	433	23.91%
Staff 20	9	2	22.22%
Staff 21	10	2	20.00%
Staff 22	193	35	18.13%
Staff 23	817	148	18.12%
Staff 24	2,412	426	17.66%
Staff 05	2,806	494	17.61%

Staff 14 (n=6), Staff 19 (n=4), Staff 20 (n=9), and Staff 21 (n=10) have too little volume for their percentage to be meaningful - flagged here only because raw volume happened to surface them, not as findings in their own right. Staff 03, Staff 09, Staff 16, Staff 04, Staff 12, Staff 08, Staff 18, and Staff 07 all have n > 700 and are the entries worth acting on.

This is not a physician-level denominator. Staff 03, Staff 09, Staff 04, and Staff 08 (Table 17) do not appear on the physician tables (Tables 7, 10) because entrants are predominantly nursing/MA staff, not the physician of record. Answering "what share of Physician 12's or Physician 02's own total BP volume gets edited" - raised directly in review - requires a join from this denominator to Enc_Provider that has not yet been obtained; see Limitations.

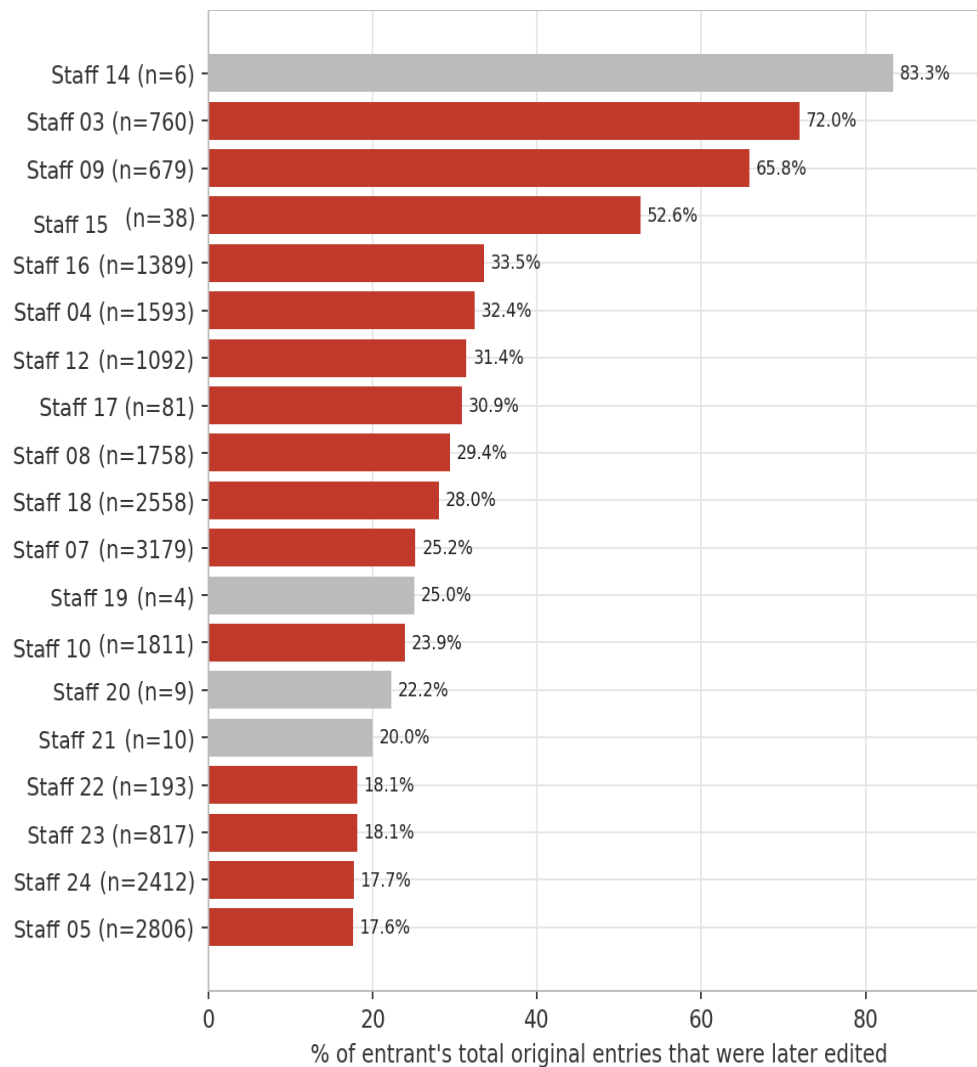


Figure 11. Entrant-level edited share. Grey bars mark entrants with fewer than 30 total entries - directional only, not reliable percentages.

3.13 Age-Cohort Deep Dive: Edit Magnitude as a Function of Age

Section 3.8 (Table 12) already shows edit magnitude differs significantly by age cohort. This section asks a sharper version of that question, in the same spirit as the review request to examine age, race, sex, and ethnicity as possible motivators: does the size of the edit, not just whether one happens, track age in a consistent direction, and does it interact with round-number preference and control-rate reclassification the same way?

Table 18. Age-cohort edit-magnitude deep dive (baseline for effect size = 18–39 cohort).

Age Cohort	N	Mean Δ Sys	Mean Δ Dia	Cohen's d vs. 18–39	Round % (Edited)	% Reclassified (of initially uncontrolled)
<18	205	−6.95	−1.00	0.121	22.4%	87.2%
18–39	1,805	−8.63	−4.87	0.000	28.9%	72.8%
40–59	5,348	−9.34	−5.17	−0.054	30.0%	73.3%
60–79	9,675	−12.29	−2.95	−0.267	30.3%	76.0%
80+	3,324	−13.48	−2.00	−0.321	31.7%	75.9%

Two distinct patterns emerge. Systolic edit magnitude increases monotonically with age, from -6.95 mmHg (<18) to -13.48 mmHg ($80+$), a small-to-medium effect size against the youngest adult cohort. Diastolic magnitude moves the opposite direction - largest in the $18-59$ range (-4.87 to -5.17) and smallest in patients $60+$ (-2.00 to -2.95). Clinically, this divergence is consistent with isolated systolic hypertension becoming more prevalent with age (elevated systolic pressure with normal or low diastolic), so a bigger systolic-only correction in older patients is not on its face surprising - but it is also exactly the shape a target-driven systolic-only adjustment would take, and this data cannot distinguish the two explanations. The reclassification rate differs significantly across cohorts ($\chi^2 = 18.86$, $p < 0.001$), though the <18 cohort's 87.2% figure rests on only 47 initially uncontrolled cases and should be treated as directional. Round-number share on edited values also differs significantly across cohorts ($\chi^2 = 11.03$, $p = 0.026$), rising from 22.4% (<18) to 31.7% ($80+$), though the range is more modest than the systolic-magnitude gradient.

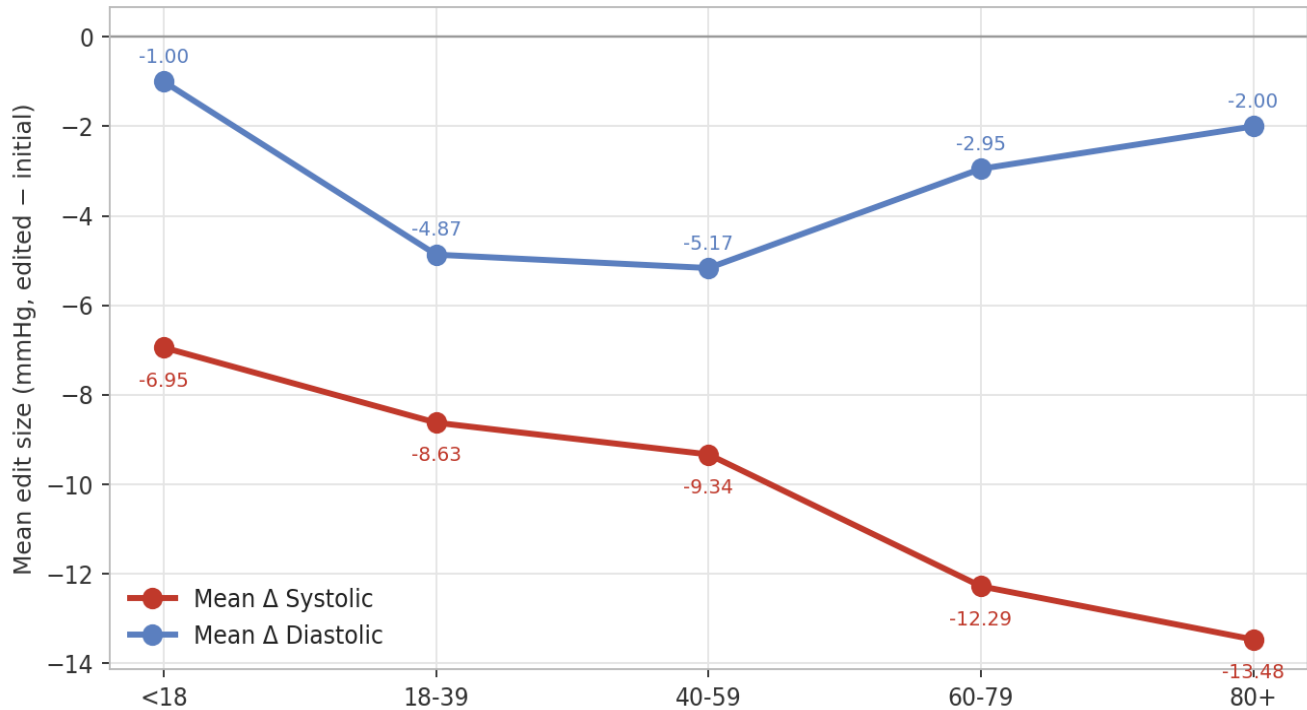


Figure 12. Mean systolic and diastolic edit size by age cohort. Systolic magnitude grows with age; diastolic magnitude shrinks - a diverging, not uniform, pattern.

3.14 Sex/Race/Ethnicity Deep Dive: Edit Magnitude

The equivalent magnitude-based deep dive for sex, race, and ethnicity - does edit size, not just edited share, differ by these factors - mirrors Table 18's structure (mean Δ by group, Cohen's d against the largest group, ANOVA/Levene across groups, digit-preference chi-square, reclassification rate), restricted to groups with at least 30 edited encounters so small-N noise doesn't drive the comparison.

Table 19. Sex-level edit-magnitude deep dive (baseline for effect size = Female, the larger group).

Sex	N	Mean Δ Sys	Mean Δ Dia	Cohen's d vs. Female
Female	11,643	-11.54	-3.12	0.000
Male	8,668	-11.04	-4.09	0.036

Sex shows the same pattern the Section 3.11 edited-share table hinted at: Female encounters are both somewhat more likely to be edited at all (0.28% vs. 0.22%, Table 14) and, once edited, show a larger mean systolic shift (-11.54 vs. -11.04). The magnitude difference is small (Cohen's d = 0.036, well below a meaningful-effect threshold) but the ANOVA is statistically significant given the sample size ($F = 6.37$, $p = 0.012$). More notable is Levene's test: $F = 44.35$, $p < 0.001$ - the *spread* of edit sizes differs sharply by sex, meaning Male edits are more variable (a mix of very small and very large changes) even though the average is similar to Female's.

Table 20. Race-level edit-magnitude deep dive, groups with $n \geq 30$ (baseline for effect size = White, the largest group).

Race	N	Mean Δ Sys	Mean Δ Dia	Cohen's d vs. White
White	14,850	-11.42	-3.40	0.000
Black or African American	4,590	-11.10	-3.94	0.023
Other	419	-11.22	-3.52	0.014
Decline to Answer	132	-11.10	-3.44	0.023
Other Asian	107	-10.93	-2.63	0.035
Asian Indian	70	-10.71	-4.01	0.050
American Indian	31	-14.77	-3.06	-0.235

Table 21. Ethnicity-level edit-magnitude deep dive, groups with $n \geq 30$ (baseline for effect size = "Not Hispanic, Latino/a, or Spanish origin," the largest group).

Ethnicity	N	Mean Δ Sys	Mean Δ Dia	Cohen's d vs. baseline
Not Hispanic, Latino/a, or Spanish origin	18,898	-11.34	-3.53	0.000
Hispanic or Latino	590	-11.03	-2.99	0.022
Unknown	475	-11.65	-4.53	-0.022
Decline to Answer	185	-10.91	-3.13	0.030
Other Hispanic, Latino/a, or Spanish origin	107	-11.39	-3.03	-0.004
Mexican, Mexican American, or Chicano/a	31	-6.84	-3.32	0.319

Race and ethnicity, unlike sex and age, do not show a system-wide magnitude signal at the major-group level: the overall ANOVA is not significant for either race ($F = 0.67$, $p = 0.67$) or ethnicity ($F = 0.76$, $p = 0.58$), and neither is Levene's test for ethnicity ($p = 0.28$), though it is marginal for race ($p = 0.019$). The two largest race groups, White and Black or

African American, sit within half a point of each other on mean systolic change (−11.42 vs. −11.10) - a materially different picture than Section 3.11's edited-share table, where both groups also sat close together (0.25% each), together indicating race is not a driver of edit *behavior* in this dataset even though it was worth checking directly rather than assuming. Two small groups depart from the pattern in opposite directions - American Indian (n = 31, d = −0.235, larger-than-average edits) and Mexican/Mexican American/Chicano/a (n = 31, d = 0.319, smaller-than-average edits) - but at n = 31 each, both are directional observations, not findings to act on without more volume.

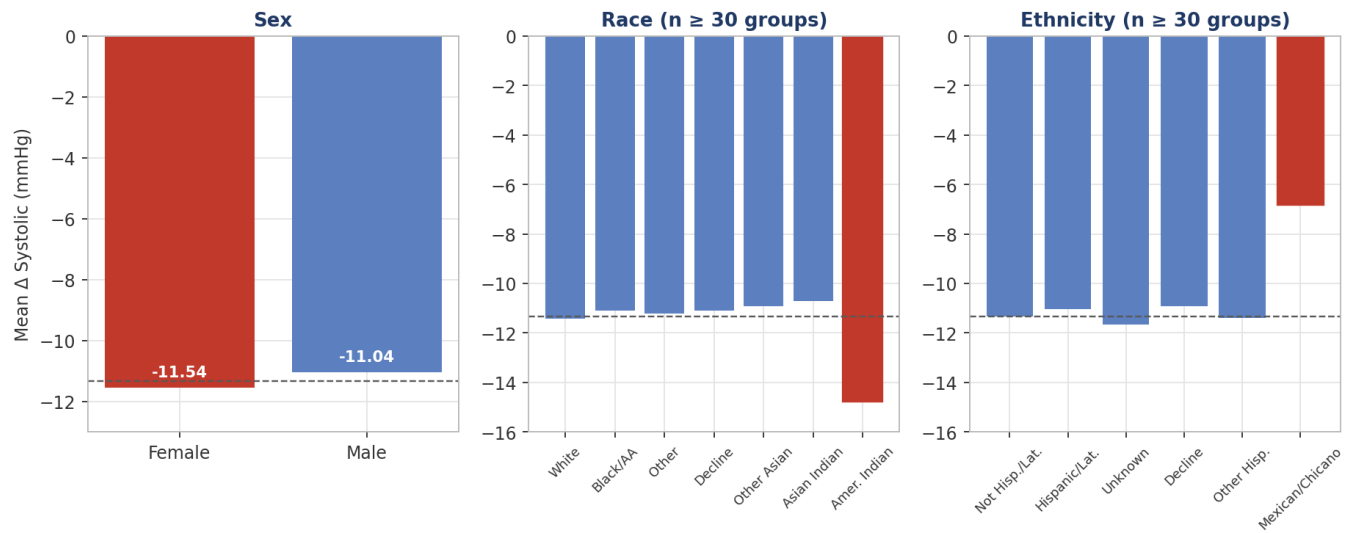


Figure 13. Mean systolic edit size by sex, race, and ethnicity (n ≥ 30 groups). Dashed line marks the system-wide mean, −11.33 mmHg. Red bars are the groups furthest from that line.

3.15 Timing Deep Dive: Does When an Edit Is Filed Predict Its Size?

Section 3.9's Q4-concentration finding (28.4% of edits fall in Q4 vs. an even 25%, and specific physicians running well above that) was raised as an open question in review: is editing done consistently throughout the year, or concentrated when it counts? Volume answers half of that question; this section asks the other half - is the average edit bigger near year-end, or just more frequent - and separately tests Physician 24's concern that an edit filed long after the original reading reflects a different, more suspect workflow than a same-visit correction.

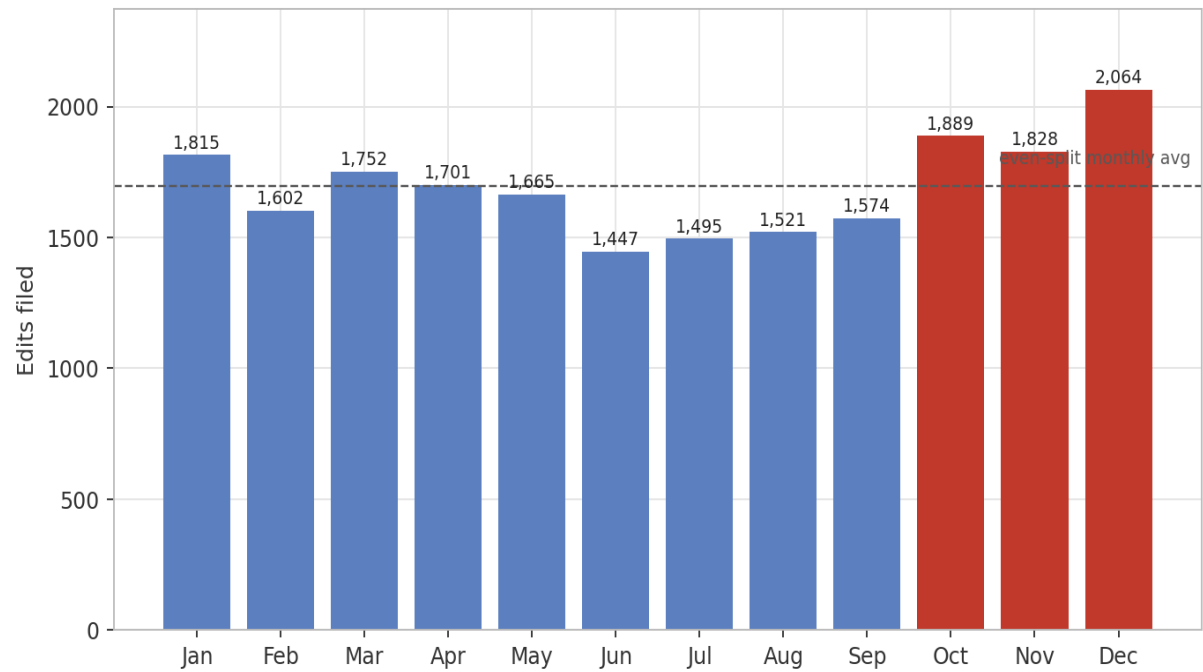


Figure 14. Monthly edit volume, CY2025. Q4 months (red) carry 28.4% of all edits against an even 25% split.

Table 22. Filing-delay distribution (time between original reading and the edit being filed).

Delay	N	% of Edits
< 1 hour	17,919	88.02%
1 hour – 1 day	1,938	9.52%
1 – 7 days	428	2.10%
7 – 30 days	38	0.19%
30 – 90 days	23	0.11%
90+ days	11	0.05%

88.0% of edits are filed within an hour of the original reading, and 97.5% within a day. Only 34 edits (0.17%) are filed more than 30 days later, and 11 (0.05%) more than 90 days later - the kind of gap Physician 24 flagged as clinically hard to justify. Filing delay has essentially no relationship to edit size or direction: Pearson $r = 0.0009$ ($p = 0.90$) between delay and signed systolic change, and $r = -0.0064$ ($p = 0.36$) between delay and edit magnitude. A direct comparison of same-day/next-day edits ($n = 19,857$, mean $\Delta = -11.25$) against edits filed 7+ days later ($n = 72$, mean $\Delta = -10.29$) shows no significant difference (Welch $t = -0.67$, $p = 0.51$). Late edits, though rare, do not look systematically different from same-day edits in this dataset.

On the year-end question: system-wide, the mean Q4 systolic edit (-10.99 mmHg, $n = 5,781$) is actually slightly smaller in magnitude than the Q1–Q3 mean (-11.46 mmHg, $n = 14,576$) - a small but statistically significant difference in the

opposite direction from a "harder push at year-end" narrative (Welch $t = 2.19$, $p = 0.029$). Per-physician, none of the individually flagged physicians show a statistically significant Q4-vs-rest difference in edit size:

Table 23. Per-physician Q4 vs. rest-of-year edit magnitude (volume concentration already reported per Section 3.9/Table 12's discussion; this table is size, not volume).

Physician	N	N in Q4	Mean Δ Sys, Q4	Mean Δ Sys, Rest of Year	t	p
Physician 12	343	78	-18.74	-19.95	0.639	0.524
Physician 02	285	98	-10.65	-9.27	-1.033	0.303
Physician 05	1,156	387	-19.82	-19.45	-0.426	0.670
Physician 08	565	133	-11.52	-12.29	0.720	0.472
Physician 06	487	19	-11.00	-17.84	1.716	0.103
Physician 04	414	6	-20.67	-21.01	0.035	0.973
Physician 01	137	61	-22.02	-20.13	-0.822	0.413

Read together with the Section 3.9 volume finding: several physicians file a disproportionate share of their edits in Q4, but the individual edits they file then are not measurably bigger or smaller than the ones they file the rest of the year. Volume concentrates near year-end for some physicians; magnitude does not. This is consistent with the review observation that the Q4 push did not actually improve Physician 12's control-rate numbers (he decreased in December) - more editing activity, not a different kind of editing.

A month-end clustering check (last 3 calendar days of the month vs. the rest) found no meaningful clustering: 9.6% of edits fall in the last 3 days of a month, against an expected 9.9% if edits were spread evenly, and the mean edit size in that window (-11.42) is not different from the rest of the month (-11.32, Welch $t = -0.29$, $p = 0.77$). Weekday volume is heavily concentrated Monday-Friday, as expected for an ambulatory clinic schedule, and is not treated here as a suspicious pattern on its own.

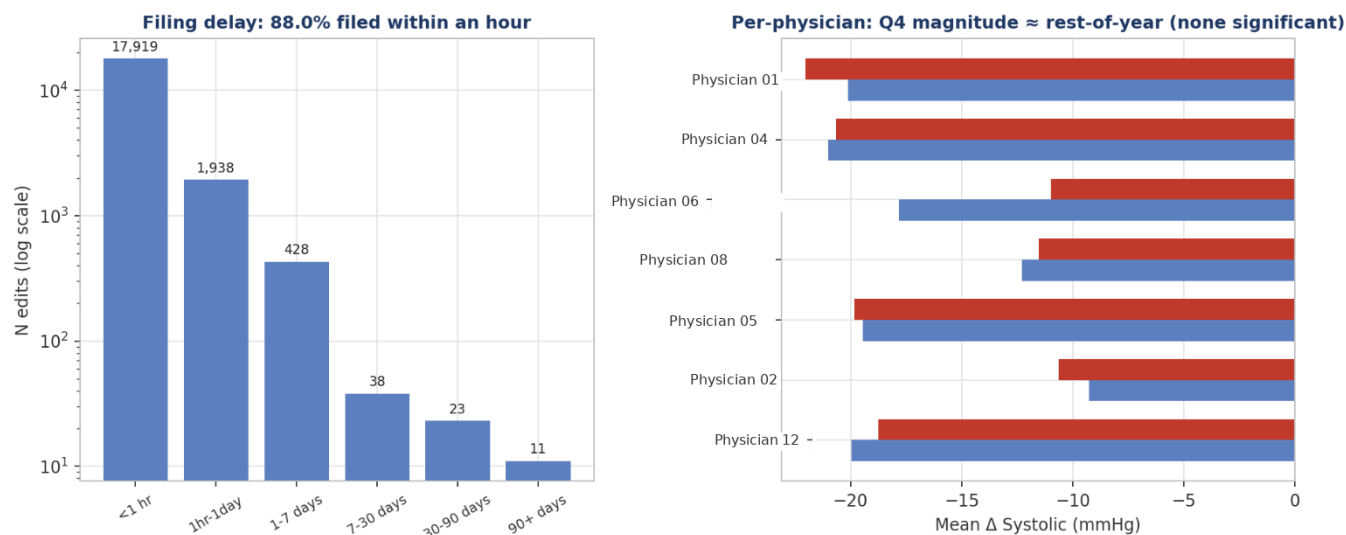


Figure 15. Left: filing-delay distribution (log scale) - the large majority of edits are filed within an hour. Right: per-physician Q4 vs. rest-of-year edit magnitude - bars track closely for every physician shown, none significantly different.

3.16 Q4 Net Benefit to the BP-Control Quality-Measure Score

Section 3.15 establishes that Q4 edits are not individually bigger than edits filed the rest of the year - volume concentrates near year-end for some physicians, magnitude does not. That leaves open the sharper question raised in review: even at ordinary size, do Q4 edits move the needle on the reported quality measure more than an ordinary edit does, and for whom? This section applies the same ≥ 140 systolic / ≥ 90 diastolic "uncontrolled" threshold used in Section 3.2 directly to each edit's initial and edited values, per physician of record (Enc_Provider), and classifies every edit as a benefit (reclassifies an encounter from uncontrolled to controlled), a cost (controlled to uncontrolled), or neutral. Net benefit is benefit count minus cost count - a direct measure of how many score-relevant reclassifications a physician's Q4 edits are responsible for - compared against that same physician's own rest-of-year rate with a chi-square test, so the comparison is each physician against their own baseline, not against each other.

System-wide, Q4 edits produced a net benefit of 3,446 reclassifications to "controlled" out of 5,781 Q4 edits - a 59.6% net-benefit rate - compared with 8,388 net of 14,576 edits (57.5%) for the rest of the year. The two rates are close enough that Q4 is not, in aggregate, a dramatically different regime system-wide; the signal is concentrated in a specific subset of physicians, not spread evenly across all 271.

Table 24. Physicians with the largest raw Q4 net benefit (net score-relevant reclassifications from Q4 edits alone). Physicians whose Q4 rate is also statistically different from their own rest-of-year baseline (Table 25) are marked.

Physician	N Q4 Edits	Net Benefit, Q4	Net Benefit Rate, Q4	Rest-of-Year Rate	Sig. Different?
Physician 05	387	308	79.6%	73.6%	Yes (p = 0.017)
Physician 23	235	224	95.3%	97.4%	No
Physician 22	261	221	84.7%	83.8%	No
Physician 19	455	209	45.9%	39.4%	Yes (p = 0.041)
Physician 08	133	126	94.7%	93.5%	No
Physician 25	126	122	96.8%	86.2%	Yes (p = 0.003)
Physician 07	110	107	97.3%	90.0%	Yes (p = 0.028)
Physician 21	165	100	60.6%	66.5%	No
Physician 26	101	98	97.0%	89.7%	Yes (p = 0.034)
Physician 27	91	91	100.0%	92.1%	Yes (p = 0.008)
Physician 17	104	87	83.7%	84.8%	No
Physician 28	97	84	86.6%	84.3%	No
Physician 02	98	83	84.7%	84.0%	No
Physician 09	91	75	82.4%	83.1%	No
Physician 15	78	71	91.0%	77.5%	Yes (p = 0.013)

Ranked by raw Net Benefit, Q4 (largest absolute effect on reported control rate). Full 271-physician table available on request.

Of the 271 physicians with Q4 edit volume, 21 show a Q4 net-benefit rate that is statistically different from their own rest-of-year baseline (chi-square $p < 0.05$, both windows $n \geq 10$ to avoid an unreliable small-sample test). Twenty of the twenty-one move in the same direction - a higher net-benefit rate in Q4 than the rest of the year - with effect sizes ranging from a modest few points (Physician 26, 97.0% vs. 89.7%) to a large jump (Physician 29, 86.7% vs. 31.0%, $p < 0.001$). One physician, Physician 06, moves in the opposite direction: a 47.4% net-benefit rate on 19 Q4 edits against an 84.8% rest-of-year baseline on 468 edits ($p = 1.4 \times 10^{-5}$) - a real and statistically significant decline in Q4, not an artifact of small Q4 volume alone, since the same physician's non-Q4 pattern is both higher and built on a much larger base.

Table 25. All physicians whose Q4 net-benefit rate is statistically different from their own rest-of-year baseline (chi-square $p < 0.05$, both windows $n \geq 10$), ranked by raw Q4 net benefit.

Physician	N Q4 Edits	Net Benefit, Q4	Q4 Rate	Rest-of-Year Rate	p
Physician 05	387	308	79.6%	73.6%	0.017
Physician 19	455	209	45.9%	39.4%	0.041
Physician 25	126	122	96.8%	86.2%	0.003
Physician 07	110	107	97.3%	90.0%	0.028
Physician 26	101	98	97.0%	89.7%	0.034
Physician 27	91	91	100.0%	92.1%	0.008
Physician 15	78	71	91.0%	77.5%	0.013
Physician 20	132	56	42.4%	57.1%	0.012
Physician 30	62	53	85.5%	51.6%	< 0.001
Physician 03	55	49	89.1%	74.0%	0.022
Physician 31	38	32	84.2%	64.2%	0.022
Physician 29	30	26	86.7%	31.0%	< 0.001
Physician 32	22	20	90.9%	66.7%	0.047
Physician 33	21	18	85.7%	40.9%	< 0.001
Physician 34	23	16	69.6%	34.1%	0.002
Physician 35	14	12	85.7%	41.2%	0.011
Physician 36	21	11	52.4%	9.1%	< 0.001
Physician 06	19	9	47.4%	84.8%	< 0.001
Physician 37	13	7	53.8%	25.0%	0.030
Physician 38	10	5	50.0%	14.3%	0.019
Physician 39	19	0	0.0%	32.3%	0.025

Physician 20, Physician 06, and Physician 39 are the only three of the 21 whose Q4 rate is lower, not higher, than their own baseline.

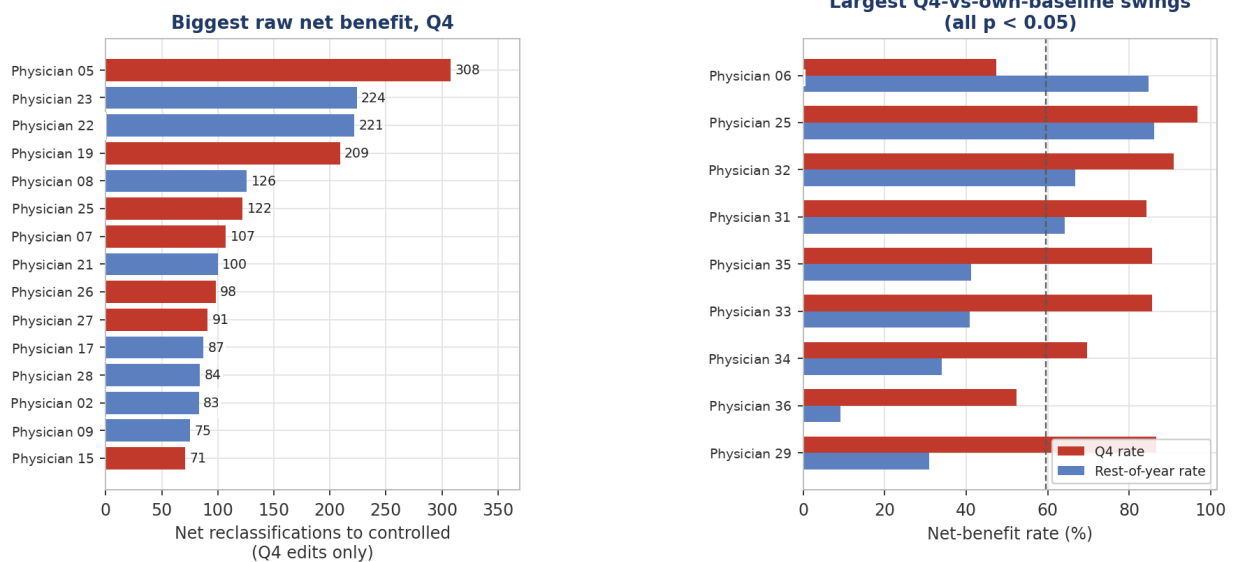


Figure 16. Left: physicians with the largest raw Q4 net benefit; red bars are statistically different from that physician's own rest-of-year baseline. Right: Q4 rate vs. rest-of-year rate for the largest statistically significant swings - every bar pair moves in the same direction except Physician 06, the one physician whose Q4 rate is lower than their own baseline.

This metric measures an outcome of edits already made - it is not, on its own, evidence of intent. A high net-benefit rate is the expected and desired result of a legitimate correction to a genuinely mismeasured or mistyped uncontrolled reading, and most physicians in Table 24 also appear in the flagged Table 7/Table 10 volume lists, so this is best read as one more lens on physicians already identified for chart-level review, not a new, independent list. Physician 06 is the one case in this section that cuts against every other finding in this report - a below-baseline Q4 net-benefit rate is not the pattern a target-driven edit would produce and may be worth chart-level review as a genuine negative control alongside the physicians flagged for the opposite reason.

3.17 Provider-Level Averages, With vs. Without Edits - Two Ways

Every prior section measures the edit itself - how big it is, who made it, when it was filed, whether it crosses the control threshold. This section asks the question the way a quality dashboard would show it: what does a physician's reported average systolic/diastolic look like using the initial (pre-edit) value versus the edited (post-edit) value. This turns out to have two legitimate but very different answers depending on what population the average is taken over, so both are presented here rather than picking one: first, the view already used elsewhere in this report - the edited encounters' own before/after average - and second, a corrected view that dilutes that same effect across each physician's, department's, and entrant's total BP volume, not just the encounters that were edited. Neither view is wrong; they answer different questions and showing both is more honest than either alone.

3.17.1 View One - Within the Edited Encounters Only (as filed elsewhere in this report)

This is the same lens used in Sections 3.13–3.16: among a physician's edited encounters specifically, what is the average of the initial values versus the average of the edited values, tracked as a running year-to-date (YTD) figure through the year. This view answers the question, "when a physician's readings do get edited, how different do the edited values look from the originals?" which is useful for flagging where editing behavior itself looks unusual but does not describe the physician's overall reported average. This is because it excludes every reading that was never touched. This is limited, as throughout Sections 3.1–3.8, to the 20,357 edited encounters.

System-wide, the gap between the YTD initial average and the YTD edited average is essentially flat all year: 12.0 mmHg in January, 11.3 mmHg by December. Editing consistently shifts the average of edited encounters down by roughly the same amount in every month - there is no month where the gap widens or narrows meaningfully, which argues against a pattern that tightens specifically around a reporting deadline and is consistent with Section 3.15's finding that edit size does not concentrate near year-end.

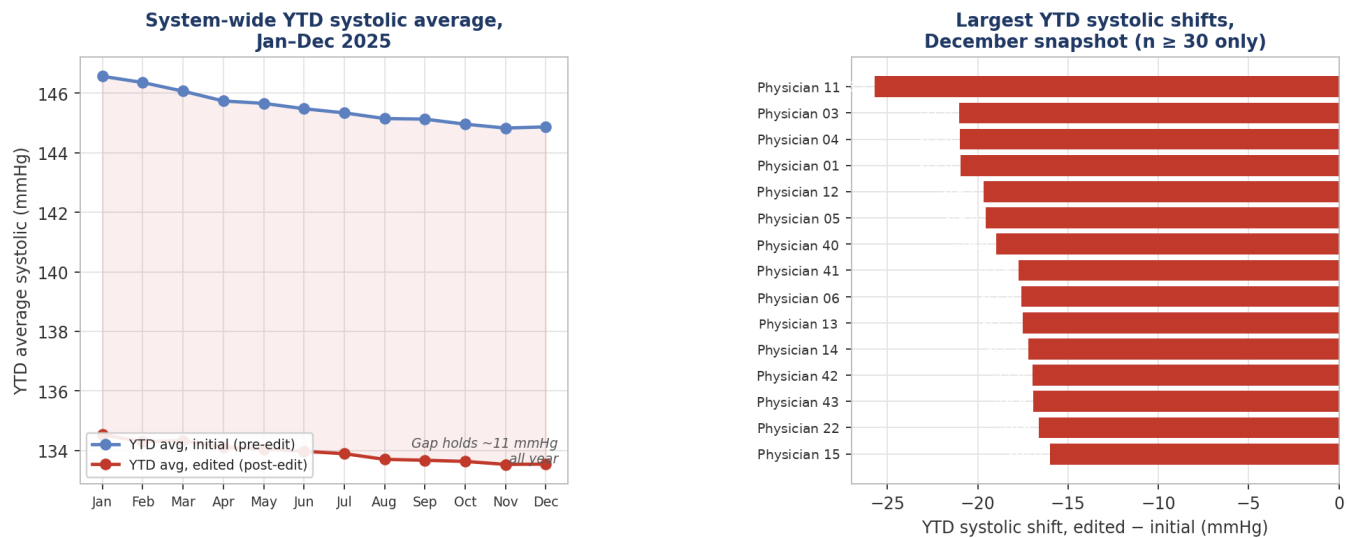


Figure 17. Left: system-wide YTD average systolic, initial vs. edited, Jan–Dec 2025 - the roughly 11 mmHg gap holds steady all year. Right: physicians with the largest December YTD systolic shift, filtered to $n \geq 30$ edited encounters for the year so the list isn't dominated by 1–6-encounter outliers.

Table 26. Physicians with the largest December YTD systolic shift (edited average – initial average, cumulative Jan–Dec), filtered to physicians with at least 30 edited encounters for the year - 109 of 271 physicians meet that bar.

Physician	N (YTD)	YTD Avg Sys, Initial	YTD Avg Sys, Edited	YTD Δ Sys	YTD Δ Dia
Physician 11	113	161.07	135.35	–25.72	–3.20
Physician 03	105	151.01	129.99	–21.02	–5.78
Physician 04	414	151.94	130.94	–21.00	–5.27
Physician 01	137	147.98	127.01	–20.97	–7.13
Physician 12	343	152.05	132.38	–19.67	–5.20
Physician 05	1,152	153.61	134.06	–19.55	–5.23
Physician 40	31	164.03	145.03	–19.00	–8.38
Physician 41	516	153.73	135.97	–17.76	0.17
Physician 06	487	149.47	131.90	–17.57	–5.83
Physician 13	170	146.57	129.07	–17.50	–4.00
Physician 14	59	155.39	138.19	–17.20	–5.59
Physician 42	98	158.04	141.06	–16.98	–2.24
Physician 43	262	154.87	137.94	–16.93	–2.75
Physician 22	1,028	149.85	133.22	–16.63	–1.99
Physician 15	207	149.64	133.63	–16.01	–3.54

Full 109-physician ($n \geq 30$) table available on request; 162 additional physicians fall below the $n = 30$ floor and are excluded here as directional-only.

Nine of these fifteen physicians already appear in this report's other flagged lists (Table 7 volume, Table 23 Q4 magnitude, or Table 24/25 Q4 net benefit) - Physician 12, Physician 05, Physician 06, Physician 22, Physician 41, Physician 01, and Physician 04 among them - so this view serves as confirmation rather than discovery of a new set of names. Physician 11 and Physician 03 are the two names in the top five that have not previously stood out in this report; Physician 11's 113 edited encounters and 25.7 mmHg YTD gap is large enough at that volume to warrant possible chart-level review previously recommended to the other flagged physicians, not dismissed as a small-sample artifact.

3.17.2 View Two - Diluted Across Each Scope's Total BP Volume (corrected for population size)

View One is accurate about the edit itself, but taken alone it invites a misreading: it looks at edited encounters in isolation, so an 11 mmHg average shift within that slice can read as if it describes a physician's or department's overall reported numbers, when in fact edited encounters are a small fraction of total volume (0.25% system-wide, per Section 3.10). This view corrects for that by diluting the same underlying edit data across the true total BP volume for each scope - every reading, edited and unedited - which is the number that actually describes what changed in the organization's real reported average. The two are mathematically consistent since, as unedited readings are unchanged in both the "with edits" and "without edits" scenario, they cancel out of the comparison and the population-level shift reduces to $\text{sum}(\text{edited value} - \text{initial value})$ across edited encounters, divided by that scope's total BP volume, computed from the full CY2025 denominator extract (Section 2.4).

Table 27. System-wide average shift, shown both ways - the same underlying edits, two denominators.

View	Population	Mean Systolic Shift	Mean Diastolic Shift
View One - edited encounters only	20,357 edited encounters	-11.33 mmHg	-3.53 mmHg
View Two - whole population, diluted	8,233,725 total CY2025 BP entries	-0.028 mmHg	-0.009 mmHg

Both numbers describe the same edits. View One describes the edit itself; View Two describes its effect on the organization's actual reported average once diluted by everything that was never touched. Neither supersedes the other - they answer different questions.

At system level, the whole-population-diluted effect is close to zero: -0.028 mmHg systolic against a typical reading in the 140s, an effect no one would notice in an aggregate report. The picture changes at smaller scopes, where edited encounters make up a much larger share of that scope's own volume - this is the same volume-concentration finding as Section 3.10 (department edited share ranges from ~2% to 11.1%), now expressed as its effect on the average rather than as a share of encounters.

Table 28. Departments with the largest whole-population-diluted systolic shift (all CY2025 BP entries in that department, edited and unedited).

Department	N Edited	N Total (dept.)	Pop.-Diluted Shift, Sys	Pop.-Diluted Shift, Dia
Department 04	2,349	21,135	-1.94	-0.49
Department 02	757	8,184	-1.89	-0.49
Department 09	2,214	21,544	-1.69	-0.19
Department 07	552	7,477	-0.99	-0.27
Department 05	2,244	20,778	-0.94	-0.76
Department 06	1,080	16,660	-0.76	-0.32
Department 13	346	3,361	-0.67	-1.03
Department 11	792	17,302	-0.55	-0.07
Department 01	1,017	17,136	-0.47	-0.20
Department 17	737	25,775	-0.43	-0.06
Department 08	238	10,412	-0.38	-0.10
Department 27	234	9,102	-0.35	-0.03
Department 21	1,017	37,516	-0.33	-0.04
Department 18	506	17,995	-0.29	-0.07
Department 03	577	19,885	-0.24	-0.10

Sorted by magnitude of population-diluted systolic shift, largest first. Department 04 and Department 02 - the two highest-edited-share departments in Table 13 - are also the two largest here, consistent with volume concentration driving the population-level effect.

At entrant level (the person who entered the original reading, mostly nursing/MA staff or the device interface, not physicians), the diluted effect is largest exactly where edited encounters make up the biggest share of that person's own total entries - Staff 09's 447 edited entries out of 679 total (65.8%) produce a -13.6 mmHg population-diluted shift on her own reported volume, the largest of any entrant or department in this report.

Table 29. Entrants (original-reading enterers) with the largest whole-population-diluted systolic shift on their own total entered volume.

Entrant	N Edited	N Total (entrant)	Pop.-Diluted Shift, Sys	Pop.-Diluted Shift, Dia
Staff 09	447	679	-13.62	-3.71
Staff 03	547	760	-7.41	-6.41
Staff 16	466	1,389	-6.31	-0.11
Staff 08	517	1,758	-5.97	-1.72
Staff 12	343	1,092	-4.55	-0.50
Staff 07	801	3,179	-4.36	-0.50
Staff 18	717	2,558	-3.44	-0.38
Staff 10	433	1,811	-3.21	-0.31
Staff 25	142	1,008	-2.93	-0.97
Staff 24	426	2,412	-2.74	-0.73
Staff 11	358	3,119	-2.43	-0.55
Staff 26	241	1,907	-2.39	-0.68

Restricted to entrants with N Total ≥ 100 to exclude single-encounter noise (several smaller entrants, e.g. $n = 9-38$ total, showed larger nominal shifts but are not reportable at that volume - see Section 3.12/Table 17 for the full entrant list including small-N rows).

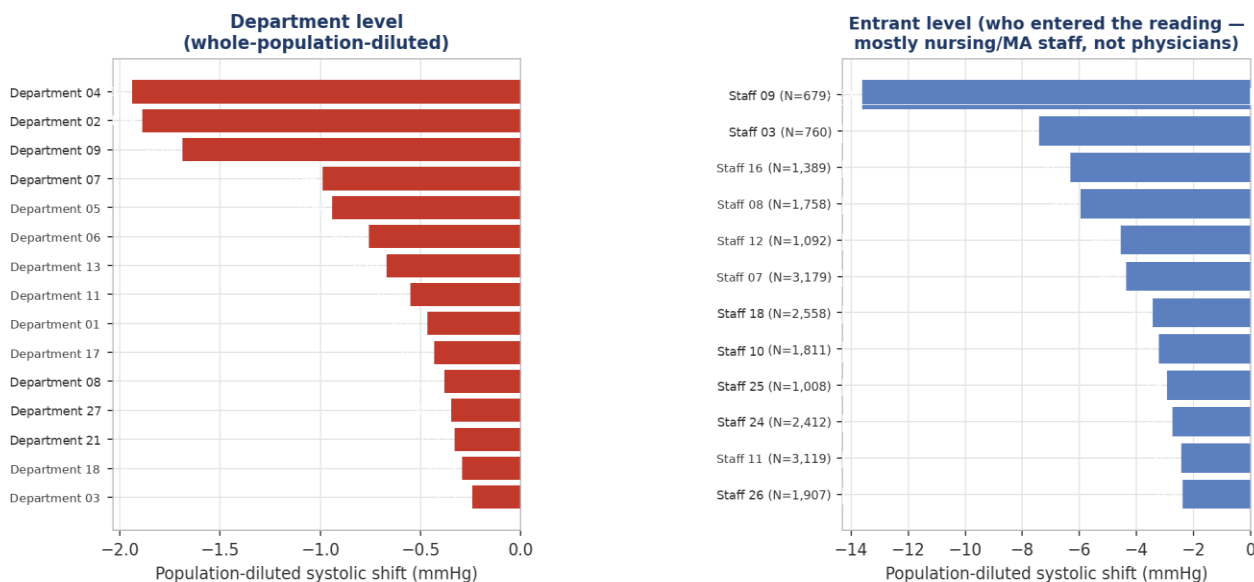


Figure 18. Left: departments with the largest whole-population-diluted systolic shift. Right: entrants (original-reading enterers, mostly nursing/MA staff) with the largest diluted shift on their own total entered volume, labeled with N.

Physician-level (Enc_Provider) View Two is not included: the denominator extract has no confirmed physician-of-record field to compute a physician's true total BP volume against. A column named Entry_Provider_ID was found on the denominator extract during this analysis and may be usable for this purpose, but its meaning has not been confirmed - it may represent the physician of record, the ordering provider, or something else entirely, and this report does not guess. Once confirmed, the department-level calculation above extends directly to physician level by substituting Enc_Provider for DEPARTMENT_NAME.

3.17.3 Entrant-Level Monthly Detail - How to Read These Charts

Table 29 (View Two) lists twelve entrants - the people who entered the original reading, mostly nursing or MA staff, not physicians - with the largest whole-population-diluted shift on their own total entered volume. That table gives one annual number per entrant; the two figures below break each of those twelve out by month for the full year, so a reader can see whether that annual number reflects a steady pattern or a handful of unusual months, and how each entrant's average sits relative to the clinical "uncontrolled" threshold used throughout this report (140 mmHg systolic, 90 mmHg diastolic - Section 3.2).

What the blue lines mean: the View One (edited-subset-only) lens applied at the entrant level and tracked monthly rather than as a single annual figure. The dashed line is the running year-to-date average of the original, pre-edit value for that entrant's edited encounters; the solid line is the same encounters' post-edit value. Both lines describe only the encounters that were edited, not this entrant's full reported volume - an entrant who enters hundreds of readings a month but only has a handful edited will show a real gap between the lines here even though, per Table 29's diluted view, the effect on their actual overall reported average may be small.

What the red lines mean: the View Two (whole-population-diluted) lens, matching each edited reading back to its specific row in the full denominator extract, month by month, rather than relying on the edited subset alone. This was computed for all twelve entrants, not just a subset - every red line on these two figures uses real output from that computation, nothing is invented or interpolated. For two of the twelve, Staff 16 and Staff 18, the reading-level match reached a reliable rate: their resulting monthly series reconcile closely with Table 29's independently-computed annual figure (Staff 16: -6.23 mmHg from the monthly series vs. -6.31 mmHg annually; Staff 18: -3.98 vs. -3.44), and the lines move smoothly month to month rather than erratically - both signs the match is trustworthy. Their panels are marked "✓ verified."

For the other ten, the reading-level match did not reach a reliable rate - in four cases (Staff 09, Staff 03, Staff 25, Staff 11) the resulting monthly shift is even the wrong sign relative to Table 29's verified annual figure, and in the rest it captures only 1-10% of the true annual effect. Rather than omit that data or present it as if it were trustworthy, those ten panels are marked "△ unreliable join," their red lines are shown faded, and the correct Table 29 annual figure is printed directly on the panel so the true number is never more than a glance away from the misleading one. This is a data-quality issue in matching individual readings to the denominator extract, not a problem with Table 29 itself - Table 29's annual figures for all twelve entrants come from a simpler, already-verified calculation (total edited-subset delta divided by total entrant volume) that does not depend on this reading-level join and is unaffected by it.

Reading the panels:

- Where the blue dashed line sits above the blue solid line, the original readings for that entrant's edited encounters averaged higher than the values they were changed to - the same downward pattern documented system-wide in Section 3.3 (H1).
- Where the black dotted 140/90 line falls inside a panel's visible range, that entrant's average crosses the clinical uncontrolled threshold.

For Staff 16 and Staff 18, the diluted (red) lines sit much closer together than the edited-subset (blue) lines and stay well clear of 140 mmHg all year - direct visual confirmation of Section 3.17.2's core point, that diluting by total volume shrinks the apparent effect: Staff 16's edited-subset gap is roughly 19 mmHg against a diluted gap under 6 mmHg; Staff 18's edited-subset gap is roughly 13 mmHg against a diluted gap under 4 mmHg.

A monthly diluted (View Two) equivalent was attempted for all twelve - matching each edited reading back to its row in the full denominator extract, month by month, rather than relying on the edited subset alone. That match was reliable for two of the twelve, Staff 16 and Staff 18, whose resulting monthly diluted series reconcile closely with Table 29's independently-computed annual figure (Staff 16: -6.23 mmHg from the monthly series vs. -6.31 mmHg annually; Staff 18: -3.98 vs. -3.44) and move smoothly month to month rather than erratically, both signs the match is trustworthy. For the other ten, the reading-level match did not reach a reliable rate and their diluted monthly lines are not shown here - their annual diluted figures in Table 29 are unaffected by this and stand as reported; only the monthly breakdown for those ten is incomplete, not the finding itself. Figures 19 and 20 below show both lenses for Staff 16 and Staff 18 (marked "✓ diluted verified") and the edited-subset lens only for the other ten.

YTD Average Systolic by Month — 12 Flagged Entrants (Table 29)
Diluted lines shown only for the 2 of 12 where the reading-level join reconciled with Table 29's verified annual figure

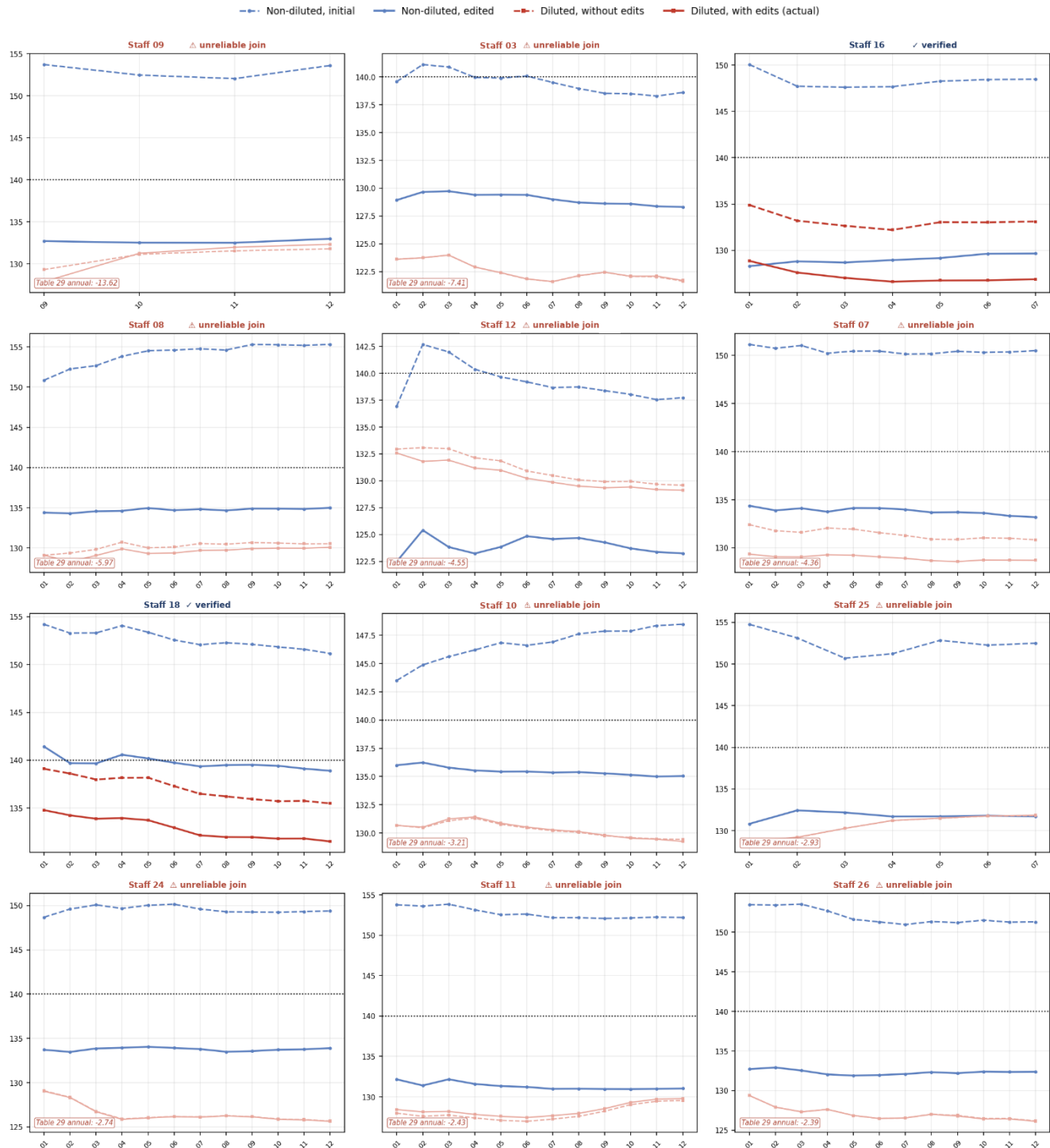


Figure 19. YTD average systolic by month for all twelve entrants in Table 29. Blue = edited-subset view (dashed = initial, solid = edited); red = whole-population-diluted view (dashed = without edits, solid = with edits/actual), computed for all twelve. Solid navy titles and full-strength red mark the two panels (Staff 16, Staff 18) where the diluted match reconciled with Table 29; faded red and an orange-flagged title mark the other ten, where the correct Table 29 annual figure is printed on the panel instead. Black dotted line = 140 mmHg uncontrolled threshold.

YTD Average Diastolic by Month — 12 Flagged Entrants (Table 29)
Diluted lines shown only for the 2 of 12 where the reading-level join reconciled with Table 29's verified annual figure

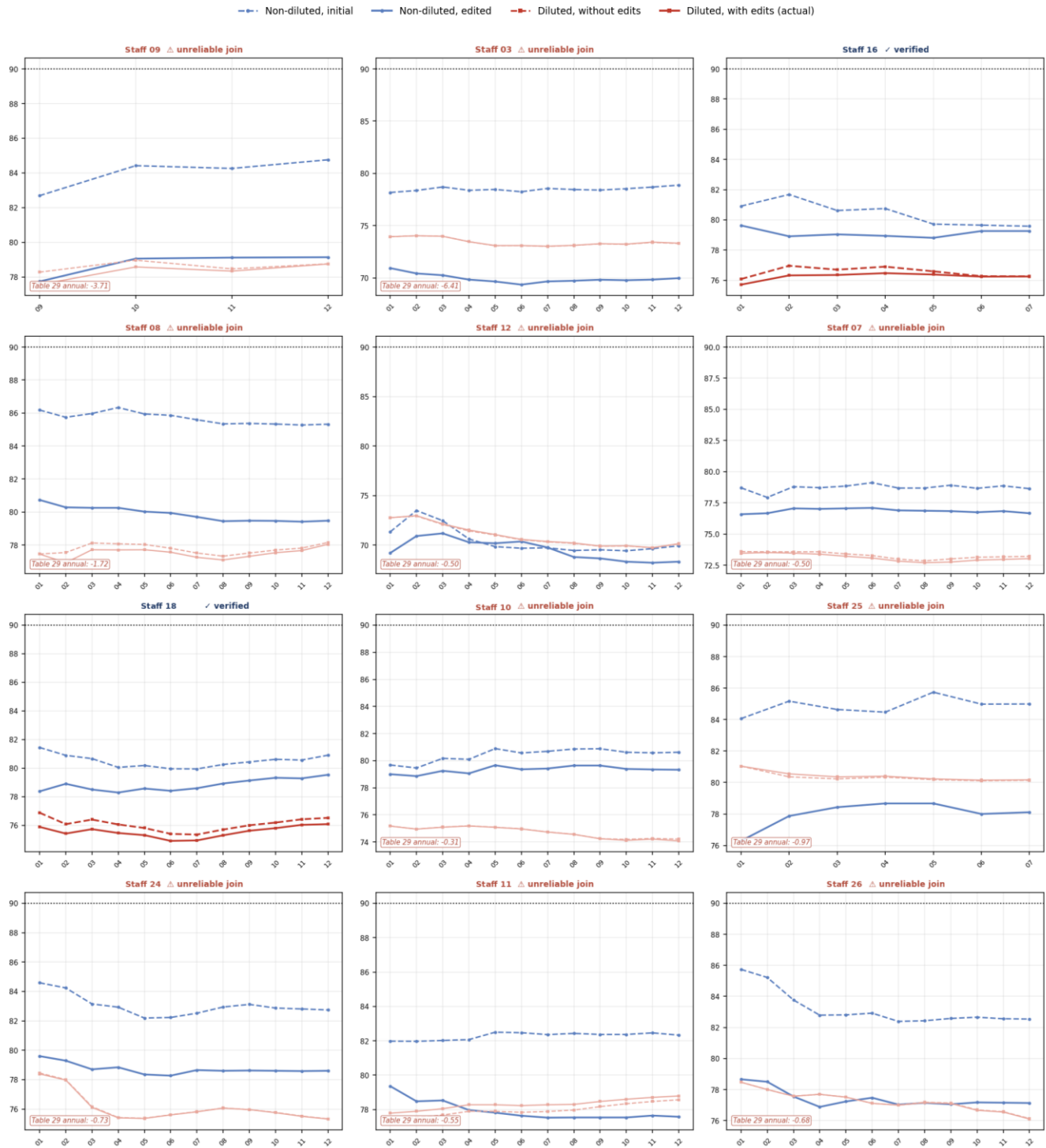


Figure 20. YTD average diastolic by month, same twelve entrants, same reading and color scheme as Figure 19; black dotted line = 90 mmHg uncontrolled threshold.

4. Limitations

- Sections 3.1–3.8 cover only encounters where a BP value was edited (20,357 records). Sections 3.9–3.17 add a full-population denominator so system, department, and demographic edited-share can now be reported directly (Section 3.10–3.12) - an earlier version of this report could not compute this and flagged it as unanswerable; that limitation is resolved as of this update. It remains true that intent behind any individual edit cannot be inferred from volume or magnitude alone.
- Physician-level edited share (what fraction of a specific physician's own total BP volume was edited - raised directly in review for Physician 12 and Physician 02) is still not available. The denominator extract identifies the entrant (Entry_User_Name, Section 3.12), not the physician of record (Enc_Provider). Obtaining this requires an additional join which can be pursued if of interest.
- The American Indian (n = 31) and Mexican/Mexican American/Chicano/a (n = 31) rows in Section 3.14 (Tables 20–21) are the smallest groups reported at the edit-magnitude level and should be treated as directional, not conclusive, until more volume can be accumulated through an expanded time frame.
- Comment-field data exists in Epic (IP_FLWSHT_EDITED.EDIT_COMMENT, IP_FLWSHT_MEAS.MEAS_COMMENT) that could, in principle, distinguish a documented recheck from an undocumented edit. That angle is not included in this version of the report pending a data-quality check on the extract.
- Section 3.17.1's (View One) $n \geq 30$ floor for the YTD-shift ranking is a volume filter, not a significance test the way Section 3.16's chi-square comparison is; it removes the 162 physicians whose annual edited-encounter count is small enough that a single unusual edit could swing their YTD average, but does not itself establish that the remaining 109 physicians' shifts are statistically meaningful.
- Section 3.17.2's (View Two) physician-level comparison is not included. The denominator extract contains a column named Entry_Provider_ID whose meaning has not been confirmed - it may be the physician of record, the ordering provider, or something unrelated - and this report does not treat an unconfirmed field as Enc_Provider. This is the same physician-level denominator gap already flagged for Physician 12 and Physician 02; confirming Entry_Provider_ID's definition would resolve it directly.
- Section 3.17.2's entrant- and department-level whole-population figures are exact given the denominator extract's counts (they use total BP volume directly, not a sample), but they inherit whatever completeness issues exist in that extract - e.g., Section 3.12 already notes 563/566 distinct entrants matched to a denominator record by name, so a small number of entrants are not represented in Table 29 at all.
- Figures 19–20 (Section 3.17.3) plot the View Two (diluted) monthly trend for all twelve Table 29 entrants, computed the same way for each - but the reading-level join underlying that computation only reached a reliable match rate for two of the twelve (Staff 16, Staff 18), confirmed by reconciliation against Table 29's independently-computed annual figure. For the other ten, the computed monthly diluted line understates the true effect (in four cases, even shows the wrong sign) and is marked accordingly on the chart with the correct Table 29 annual figure printed on the panel. Table 29's annual diluted figures for all twelve entrants do not depend on this reading-level join and are unaffected by its reliability issues.
- Section 3.16's net-benefit metric measures the outcome of edits already made against the CMS/NCQA control threshold; it does not and cannot establish intent. A high net-benefit rate is also the expected result of a correctly executed legitimate recheck of a genuinely uncontrolled reading, so Table 24/25 should be read as an additional lens on physicians already flagged by volume in Tables 7 and 10, not as an independent finding of its own.
- Intent cannot be determined from this data. A systematic downward, round-number-converging edit pattern is consistent with target-driven adjustment but this analysis cannot distinguish that from other systemic causes without chart-level review.

- Physiological-plausibility screening removed 168 of 20,532 rows using a fixed range filter; a small number of borderline-but-legitimate readings may have been excluded, and a small number of implausible-but-passing rows may remain.
- The clinical control-rate cross-check (Section 3.2, Table 3) uses a slightly different row count (20,334) than the main statistical analysis (20,357) because it applies its own independent plausibility bounds; the two are consistent in direction and magnitude but not derived from identical row sets.
- Scope is limited to Family Medicine and Internal Medicine, the only two specialties present in this extract.
- Editor-pair figures (Table 11) with lower edit counts are more sensitive to a handful of encounters and should be treated as directional below roughly $n = 100$. The same caution applies to small-N rows in Tables 15 (race) and 17 (entrant) flagged inline.
- A 2024 specialist-equipment survey (cuff type, as a possible covariate for measurement variance) was checked against this physician population and found not usable in its current state. The survey covers a different set of specialties than the Family Medicine/Internal Medicine population analyzed here.
- This report's edit-level analysis depends on the CY2025 audit extract as pulled from Clarity by the requesting analyst; if that extract systematically excludes certain edit types, departments, or time periods (for example, edits made through a workflow the extract query doesn't capture or edits that were themselves later reversed and dropped from the audit trail), the population-diluted figures in this report would understate or overstate the true effect for whatever subset is missing. This analysis cannot reliably detect such a confounding factor. The consistency of the findings across system, department, and entrant level (Sections 3.17.1–3.17.2) is reassuring but does not rule this out since a systematic extraction gap would shift all three levels together rather than showing up as an internal inconsistency.

5. Conclusion

Every level of this analysis - system, specialty, department, physician, age cohort, and editor - continues to point to the same pattern documented in the prior version of this report: BP edits in CY25 are systematically lower than the original reading, more tightly clustered around round or "clean" values than original readings are, and this pattern strengthens rather than weakens when the person editing the value is not the person who took it. This update adds seven findings that sharpen, rather than change, that picture.

First, the edited population appears to represent small slice of total BP volume (0.25% system-wide) concentrated in a handful of departments.

Second, the size of an edit grows with patient age for systolic readings specifically, a pattern with a plausible clinical explanation (isolated systolic hypertension) that this data cannot separate from a target-driven one.

Third, of the four demographic factors examined as possible motivators (age, sex, race, ethnicity), only age and, to a much smaller degree, sex show a magnitude signal. Race and ethnicity do not, at the major-group level, once edited-share volume and edit size are considered separately.

Fourth, edit timing tracks volume, not size. Some physicians file disproportionately more edits near year-end, but those edits are not bigger than the ones they file the rest of the year, and a longer gap between the original reading and the edit does not predict a larger or more directional change.

Fifth, applied directly to the quality-measure threshold, Q4 edits are responsible for a modestly higher net-benefit rate system-wide (59.6% vs. 57.5%) than the rest of the year, and for 21 physicians that Q4 shift is large enough to be statistically distinct from their own baseline - 20 of them shifting toward more benefit, and one, Physician 06, shifting the opposite direction.

Sixth, viewed as a running YTD average the way a quality dashboard would show it (Section 3.17.1), the gap between a physician's pre-edit and post-edit average - measured within edited encounters only - holds steady across the year rather than widening toward year-end (12.0 mmHg in January, 11.3 mmHg in December, system-wide); among physicians with enough volume to trust ($n \geq 30$ edited encounters), the largest December YTD shifts mostly confirm physicians already flagged elsewhere in this report.

Seventh, under the assumption that edit and population counts are accurate, the edited-subset shift does not meaningfully impact the control measure of physicians once diluted across each scope's true total BP volume rather than isolated to the encounters that were actually edited (Section 3.17.2). The -11.33 mmHg system-wide shift becomes -0.028 mmHg once diluted across all 8,233,725 CY2025 BP entries, a negligible effect on the organization's real reported average. The diluted effect is largest wherever edited encounters make up the biggest share of that scope's own volume: two departments and one entrant stand out on this basis. Every finding above is accurate about the edits themselves; this last one is the reminder that the edits themselves are a very small fraction of total BP volume, and both views are presented side by side in Section 3.17 rather than choosing one.